Quantum Mechanics

Leonard Susskind

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Chapter 1

Modern Physics: Quantum Mechanics

This first lecture opens in an unusual way. Instead of beginning with a formal postulate list, Leonard Susskind begins by explaining the style of the course and then turns at once to a deeper claim: quantum mechanics is not classical mechanics with a little noise stirred in. To make that point, he unfolds the subject in the order the lecture itself demands. We begin with classical randomness as a foil, then the two-slit experiment, then reversibility and the cost of looking, then Heisenberg's microscope, and only after that do we take the first abstract step toward complex vector spaces. Where board frames curated through LazyingArt LLC and Video2Book help stabilize equations or layout, they are retained as supporting evidence; the transcript remains the primary source for the sequence and force of the argument.

1.1 Why This Course Starts With Logic, Not Machinery

Susskind begins with the audience. This is not freshman service physics, and it is not a standard undergraduate or graduate course. It is a continuing-education course, and that fact shapes the method. The audience wants the basic ideas quickly, but not cheaply. We are to use equations when necessary, and when we do use them, we are to use the simplest equations that will actually do the job. He even jokes that one might call it quantum mechanics for old people, including himself. The joke matters less than the pedagogical point behind it: we are not here for ornament, but neither are we here for slogans.

That brief preface also explains why the lecture opens with logic rather than machinery. Susskind reminds us that classical mechanics came first in the larger sequence of courses, but quantum mechanics is more fundamental. Classical physics is what we get when quantum mechanics may be approximated by classical logic in the appropriate regime. So instead of starting with Schrödinger's equation or operator formalism, the lecture starts by asking what classical logic gets wrong. That is the right place to begin, because the mathematics introduced later is meant to replace a structure that has already failed.

1.2 Classical Randomness Is Not Quantum Unpredictability

The first task is to separate two ideas that are too easily conflated. Quantum mechanics certainly contains unpredictability. But that does not mean that it is simply classical mechanics plus random

disturbances. To make the distinction sharp, Susskind invents a classical foil.

Imagine an otherwise Newtonian world in which a deity occasionally gives the moon tiny random kicks. Every tenth of a second the dice are thrown; on one outcome the moon is nudged one way, on another it is nudged the other way. This world would be statistically unpredictable, but the unpredictability would be of a familiar classical kind. It would enter by occasional deviations from deterministic law.

Now ask what such randomness would do. It would not merely blur our predictions. It would also jostle conserved quantities. If the kicks are random, then the energy of the system would, step by step, randomly increase or decrease. On average it might drift slowly or perhaps not at all, but exact conservation would be gone. The very form of the randomness would show itself in the energy bookkeeping.

Quantum mechanics does not behave that way. We prepare a system with a precise energy, let it evolve, and then measure again. The later result may be unpredictable in some respects, but the energy comes back exactly the same. No trace remains of a hidden hand knocking the system this way and that. So the lecture's first lesson is already in place: quantum unpredictability is highly special. It is not what one would get by sprinkling classical randomness into Newton's laws.

1.3 The Two-Slit Experiment and the Failure of Additive Probabilities

Susskind's first great example is the two-slit experiment, because it exhibits the weirdness of quantum mechanics in the most direct and least encumbered way. Let us follow the order of the lecture carefully.

We begin with a source of very light particles—photons, electrons, neutrons, anything for which quantum effects are appreciable. They are sent through the apparatus one at a time. This matters. The lecture insists on that sparsity precisely so that no later particle can plausibly be blamed for what happens to the earlier one.

First open only one small hole. A particle passes through and hits the screen in an isolated blip. Repeat the experiment many times and those individual blips build up a smooth profile: a bright central region fading away to the sides. We may call that one-hole distribution $P_1(y)$. Close that hole, open the second one, and repeat. We obtain another profile $P_2(y)$, displaced relative to the first.

At this stage classical logic seems irresistible. If both holes are open, then a particle reaching a point y on the screen must have come either through the first hole or through the second. Therefore the probabilities ought to add:

$$P_{12}(y) = P_1(y) + P_2(y). \tag{1.1}$$

This is not the quantum rule. It is the classical expectation that Susskind wants fully on the table before overturning it.

Now do the real experiment. The particles still come one at a time. The screen still records one blip at a time. But the built-up pattern is no longer the sum of the two one-hole profiles. Instead we get an interference pattern, and at some points the probability vanishes entirely:

$$P_{12}(y) = 0 \qquad \text{even though} \qquad P_1(y) \neq 0, \quad P_2(y) \neq 0. \tag{1.2}$$

This is the shock of destructive interference. With only the first hole open, particles arrive at that point. With only the second hole open, particles also arrive at that point. Open both holes, and no particle ever arrives there.

That is the point where Susskind widens the claim. One could try to invent some elaborate mechanism, some baroque memory hidden in the apparatus, but the phenomenon is too generic for that sort of one-off repair. The difficulty is not a local detail. The difficulty is that classical statistics and classical logic are built on the wrong underlying structure. The additive rule for alternatives has simply failed.

1.4 Reversibility, Information, and the Cost of Looking

Having displayed the failure of additive probabilities, the lecture shifts to a second diagnostic: reversibility. The move is deliberate. Susskind wants a different test, one that is not tied to slit interference, and for that he first backs up to very simple discrete systems.

Consider a two-state system, a coin with states H and T. A deterministic law of motion can take the form

$$H \mapsto H, \qquad T \mapsto T, \tag{1.3}$$

or it can alternate the states,

$$H \mapsto T, \qquad T \mapsto H. \tag{1.4}$$

Now move to a three-state system with heads, tails, and feet:

$$H \mapsto T, \qquad T \mapsto F, \qquad F \mapsto H. \tag{1.5}$$

The important point is not which law we choose. The important point is that in each of these examples the evolution is exact and information-preserving. If we let the system evolve for a long time and then reverse the arrows of the update rule, it retraces its history and returns to the starting state.

This gives Susskind an operational definition of reversibility. We do not need the detailed law; we need only the possibility of reversing it. Evolve for a time t, reverse the rule, evolve for the same time t again. If the system always returns to its starting point, the dynamics is deterministic in the sense he wants.

Now add a tiny bit of classical randomness. Suppose that with some very small probability the system does not follow the prescribed update and instead does something else—for example, it stays put when the rule says move on. Over short times the reversal test may still succeed. But over sufficiently long times a fluctuation will almost certainly occur, and then running backward does not return us uniquely to the initial state. Information has been lost. That is why Susskind says that classical randomness destroys the conservation of information.

He now carries the same test over into quantum mechanics. Here the lecture momentarily wavers in its wording, but the logic is clear: for this thought experiment we no longer need two slits. One slit is enough. Send one electron through the slit. Remove the screen. Let the electron pass through, but do not detect it. Then reverse the law of physics and run the system backward for the same length of time.

What happens? If nothing has looked at the electron, nothing has recorded its location, nothing has disturbed it in the intermediate stage, then the reversal is exact. The electron retraces the reverse trajectory and goes back into the gun. That is already a remarkable statement. The apparent randomness of the one-slit deflection is not the same thing as classical random compounding of errors.

Now alter only one thing. Put in a detector and record where the electron is during the intermediate stage. Then reverse the law and run backward. In that case the exact return fails. The uncertainty does not cancel; it compounds. The reversibility test, which classical gentle measurement would have left untouched, is destroyed by quantum measurement itself.

This is the deeper transition of the lecture. Classical physics allows an arbitrarily gentle observation. One can, in principle, look at a classical system without significantly altering its subsequent behavior. Quantum mechanics does not. Determining the state of the system is never a harmless afterthought.

Susskind then returns to the two-slit experiment and states the same idea in a different form. The interference pattern is present only if nothing in the apparatus or environment records which slit the electron went through. That recorder need not be a human observer. It could be a gas molecule, another electron, or any degree of freedom that remembers afterward which path was taken. Once such path information exists in the environment, the interference pattern disappears and the answer reverts to the classical one: the probabilities add.

1.4.1 Question & Answer

Question. If the particle is deflected by the slit or plate, does the plate not recoil? And if it recoils, does that not already reveal which path the particle took?

Answer. The recoil is real. The issue is whether it is readable. Susskind's answer is qualitative and deliberately anticipates the uncertainty principle before formally deriving it. A detector that is very well localized in position must have a broad momentum distribution. So although an electron gives the detector a small kick, that kick only shifts an already broad momentum distribution by a small amount. Unless the shift is comparable to the width of the distribution, one cannot tell afterward whether the kick occurred.

The lesson is precise enough for our purposes. Recoil does not automatically provide which-path information. It provides readable information only if the shift in detector momentum is large compared with the detector's own intrinsic momentum uncertainty.

1.5 Heisenberg's Microscope and the Momentum of a Photon

At this point Susskind turns explicitly to the uncertainty principle. But again he does not present it first as a formal theorem. He stages it as a demand for physical explanation. Heisenberg may say that position and momentum cannot be simultaneously determined, but Bohr's challenge is: do not merely tell me that x and p fail to commute; show me the physics that forces this obstruction.

The physical setting is measurement by light. To locate a particle under a microscope, we scatter a photon from it, collect the scattered light, and form an image. So the next question is simple: what did Einstein, de Broglie, Bohr, and Heisenberg know about photons that makes this measurement problematic?

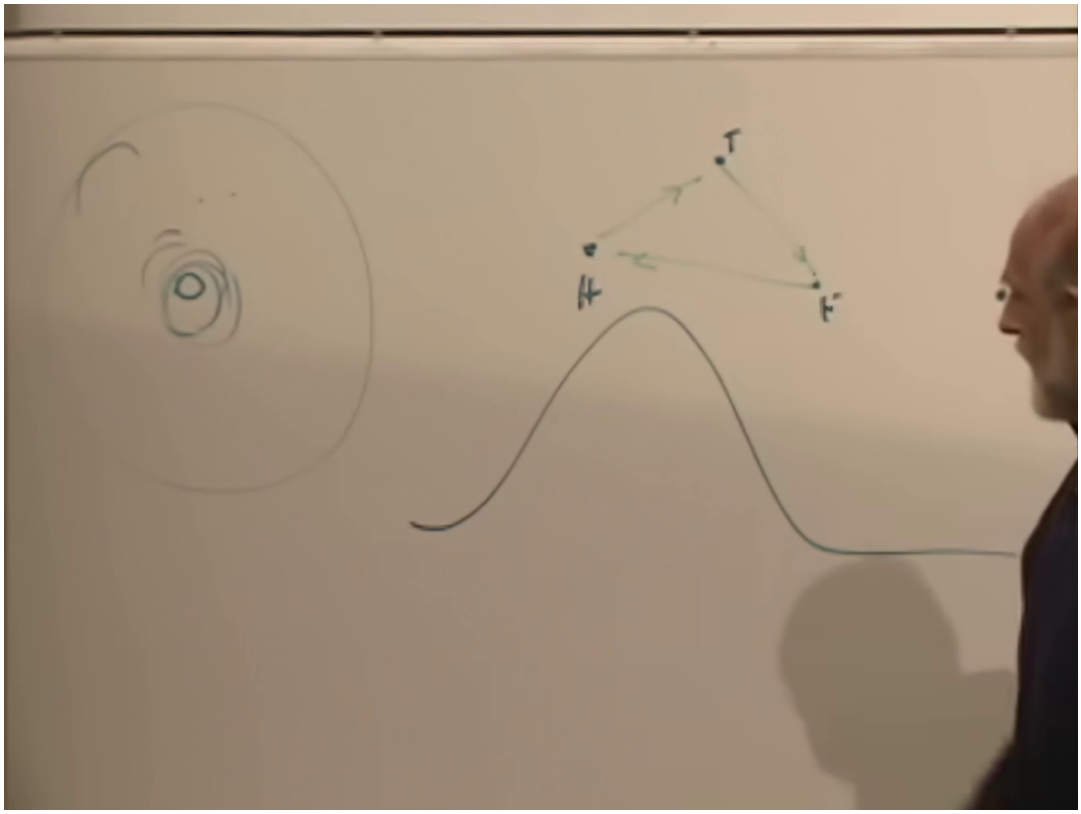


Figure 1.1: Board layout for the recoil argument. The broad central hump is the detector momentum distribution introduced in the answer; the large oval at left and the small triangle-like sketch at upper right are residual board context and should not be overread.

The board snapshot is valuable here because it preserves the cluster of relations as Susskind arranged them.

The first relation is Einstein's photon-energy formula:

$$E = hf = \hbar\omega, \tag{1.6}$$

with

$$\omega = 2\pi f, \qquad \hbar = \frac{h}{2\pi}. \tag{1.7}$$

Susskind then recalls that light carries momentum as well as energy. For ordinary nonrelativistic particles, one has

$$E = \frac{p^2}{2m} = \frac{1}{2}pv, \tag{1.8}$$

so energy and momentum already have the right dimensions to be related by a speed. For light, the relativistic relation is simpler:

$$E = cp, \qquad \text{hence} \qquad p = \frac{E}{c}. \tag{1.9}$$

Now combine this with the photon-energy formula:

$$p = \frac{hf}{c}. \tag{1.10}$$

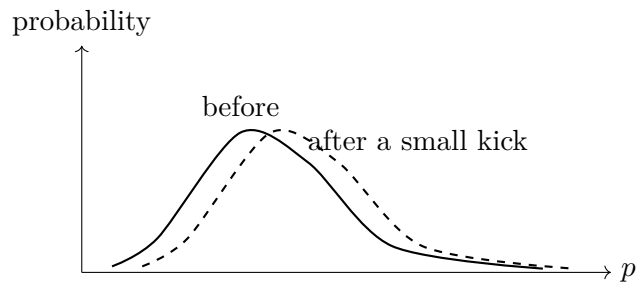


Figure 1.2: Transcript-derived redraw of the detector momentum distribution. The lecture's point is that a small recoil may shift the distribution without making that shift experimentally readable.

To eliminate the frequency we use the kinematics of a wave. If one cycle takes time T, then

$$T = \frac{1}{f}. \tag{1.11}$$

If the wavelength is λ, then in one period the wave advances by one wavelength, so

$$c = \frac{\lambda}{T} = \lambda f, \qquad \text{or equivalently} \qquad f = \frac{c}{\lambda}. \tag{1.12}$$

Substituting into the momentum relation gives the de Broglie formula:

$$p = \frac{hf}{c} = \frac{h}{c}\frac{c}{\lambda} = \frac{h}{\lambda}. \tag{1.13}$$

This is the decisive step: momentum and wavelength are inverse to one another.

The lecture then pauses and leans on the physical meaning of that inverse relation. If we want to form an image that is sharp on a scale Δx, then the wavelength used to form the image must be at most of that order:

$$\lambda \lesssim \Delta x. \tag{1.14}$$

That is not a fancy theorem here; it is ordinary imaging logic. A wavelength much larger than the desired detail necessarily produces a fuzzy image on that larger scale.

But now the trouble appears. If λ must be small, then the photon momentum must be large:

$$p_\gamma = \frac{h}{\lambda}. \tag{1.15}$$

So the very photon that makes a sharp image is also a high-momentum projectile. When it scatters from the particle whose position we are trying to determine, it gives that particle an uncontrollable momentum kick. The lecture does not derive a sharp inequality such as $\Delta x\,\Delta p \geq \hbar/2$, and we should not import one prematurely. What it does justify is the order-of-magnitude statement

$$\Delta p_{\text{kick}} \sim \frac{h}{\lambda}. \tag{1.16}$$

Thus a precise position measurement necessarily produces a large uncertainty in the post-measurement momentum.

Susskind then points out why classical physics seemed to evade this difficulty. In classical physics, light is not quantized. One imagines that one can keep the wavelength fixed while making the

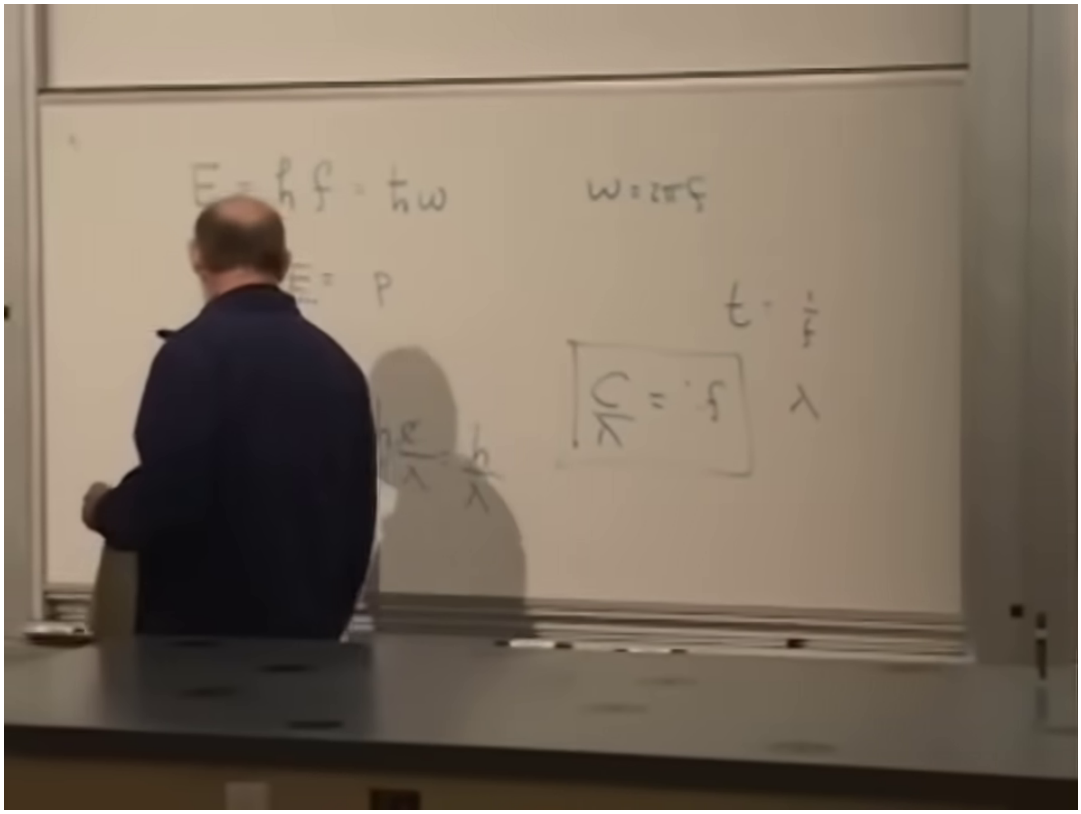


Figure 1.3: Board layout for the photon-energy and wave-kinematics relations leading to the inverse momentum-wavelength law.

intensity arbitrarily small, thereby preserving resolution without delivering a meaningful kick. Quantum mechanics blocks that maneuver. At fixed wavelength there is a minimum packet, namely one photon, and one photon already carries momentum h/λ. There is no infinitely gentle probe of a given short wavelength.

At precisely this stage of the lecture, Susskind remarks that he had erased the key formula and rewrites it. The chalk line itself is partly hidden in the frame, so the reconstruction must be guided by the transcript and the surrounding visible equations.

The physical conclusion is now inescapable. There is no perfectly gentle sharp measurement of position in quantum mechanics. If the measurement is sharp, the momentum is badly disturbed. If the measurement is gentle, the position information is necessarily coarse.

1.5.1 Question & Answer

Question. If a sharp position measurement ruins momentum, how can we measure a particle's velocity at all?

Answer. We measure it gently, and we take our time. The lecture's answer is operational. Measure the position twice, at two different times, and infer the velocity from the displacement divided by the elapsed time. But each position measurement must be gentle, so each must use a long-wavelength

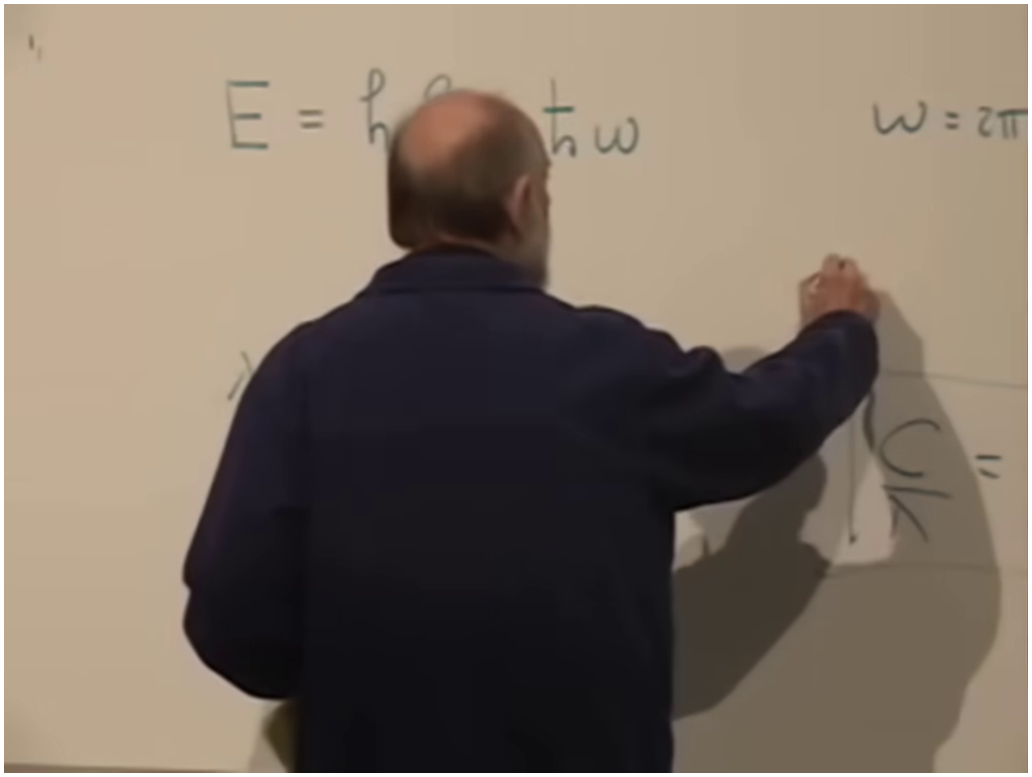


Figure 1.4: Susskind rewriting the erased momentum-wavelength relation. The exact chalk line is partly occluded, but the transcript makes clear that the restored equation is $p = h/\lambda$.

photon.

A long-wavelength photon localizes the particle only coarsely:

$$x_1 = x \pm O(\lambda), \tag{1.17}$$

$$x_2 = x + vt \pm O(\lambda). \tag{1.18}$$

Subtracting the two measurements gives the traveled distance,

$$d = x_2 - x_1 = vt \pm O(\lambda). \tag{1.19}$$

Therefore the inferred velocity has uncertainty

$$\Delta v \sim \frac{\lambda}{t}. \tag{1.20}$$

If we make t very large, then Δv can be made very small even when λ is large. So velocity may indeed be measured accurately. But the price is that the position was never known to better than order λ. This is exactly the complementary tradeoff Susskind wants us to keep in mind. We can know velocity well, but only by giving up precise knowledge of position; or we can localize sharply, but only by badly disturbing the momentum.

1.6 From Classical States to Quantum Vector Spaces

Only after all of these physical examples have done their work does the lecture make its final turn. Susskind says, in effect: now we are ready to ask what is really new in the mathematical language.

In classical mechanics, the state of a system is a point in a set. That already tells us something about the operations available. We can ask whether two sets intersect or unite, but it makes no sense to multiply a point by a number. If the only states are heads and tails, then $3 \times$ heads and $(2 + 4i) \times$ heads are meaningless expressions.

Quantum mechanics will replace this set-like notion of state by a vector-space notion. That should feel radical, and Susskind wants it to feel radical. He even says that at this point it should sound like mumbo jumbo. But the lecture is careful: before asking why physical states should be vectors, it first explains what a vector space is.

Definition 1.1. A complex vector space is a collection of objects for which two operations are defined: any two objects can be added, and any object can be multiplied by an arbitrary complex number, with the result again belonging to the same collection.

Once those operations exist, any linear combination is again an allowed object:

$$\alpha A + \beta B \in V \qquad \text{whenever} \qquad A, B \in V,\ \alpha, \beta \in \mathbb{C}. \tag{1.21}$$

That is the basic algebra the lecture wants us to absorb.

Ordinary geometric pointers provide the familiar analogy. They form a vector space over the real numbers: they may be added, and they may be multiplied by real scalars. But quantum mechanics wants something more general, namely vector spaces over the complex numbers.

The first example Susskind gives is the space of complex-valued functions of a real variable x. Here x itself is an ordinary real variable; what is complex is the value of the function:

$$\psi(x) \in \mathbb{C}. \tag{1.22}$$

If $\psi(x)$ and $\phi(x)$ are such functions, and if α is a complex number, then

$$(\alpha\psi)(x) = \alpha\,\psi(x), \tag{1.23}$$

$$(\psi + \phi)(x) = \psi(x) + \phi(x). \tag{1.24}$$

Both operations produce new complex-valued functions of x. Therefore complex functions of one real variable form a complex vector space. This is the example that will later become central in quantum mechanics.

The lecture also makes an important contrast. Real-valued functions form a vector space over the real numbers, but not over the complex numbers. The reason is immediate: multiplying a real-valued function by i does not keep it real-valued.

The second example is the fixed-length column vector. If we take a column of four arbitrary complex numbers,

$$A = \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{pmatrix}, \qquad B = \begin{pmatrix} B_1 \\ B_2 \\ B_3 \\ B_4 \end{pmatrix}, \tag{1.25}$$

then addition is defined entry by entry:

$$A + B = \begin{pmatrix} A_1 + B_1 \\ A_2 + B_2 \\ A_3 + B_3 \\ A_4 + B_4 \end{pmatrix}, \tag{1.26}$$

and multiplication by a complex scalar is equally simple:

$$\alpha A = \begin{pmatrix} \alpha A_1 \\ \alpha A_2 \\ \alpha A_3 \\ \alpha A_4 \end{pmatrix}. \tag{1.27}$$

For a fixed length, these columns form a vector space. But here the lecture is careful about a subtle point: the collection of all columns of all lengths is not a single vector space under these rules, because no addition law has been given between, say, a two-component vector and a three-component one.

Finally, Susskind steps back to the complex numbers themselves. They already form a one-dimensional complex vector space. That makes complex conjugation the next natural operation to inspect. Write

$$z = x + iy, \qquad z^* = x - iy. \tag{1.28}$$

Geometrically, z^* is the reflection of z across the real axis. Algebraically,

$$z^* z = (x + iy)(x - iy) = x^2 + y^2. \tag{1.29}$$

So the product of a number with its conjugate gives the square of its magnitude. Susskind treats this as a basic operation, not a decorative trick, because it is one of the first examples of a mapping that will later be generalized from numbers to vectors.

The closing note of the lecture is important. We are not yet being told in what precise sense physical states are vectors. That comes later. For now we are learning the abstract operations themselves. And that is exactly right: if classical logic is built on sets, then quantum logic will need a new arena, and the lecture ends by introducing the first pieces of that arena.

1.7 Summary

The lecture begins by rejecting the easy idea that quantum mechanics is just classical mechanics with random noise added. Classical random kicks would spoil exact conservation laws and destroy reversibility. Quantum mechanics does neither in that simple way. The two-slit experiment then shows that probabilities do not merely add: with both slits open, there are points where the probability vanishes even though each one-slit experiment separately gives nonzero arrivals there.

The lecture then turns the notion of reversibility into an operational test of determinism. Classical randomness destroys that test; quantum evolution preserves exact reversibility only so long as nothing records intermediate information. That is the deeper lesson behind both the one-slit reversal experiment and the two-slit which-path story: in quantum mechanics, measurement is part of the dynamics.

Heisenberg's microscope then gives the physical mechanism. To resolve position sharply we need short wavelength; short wavelength means large photon momentum; the very photon that makes the image gives the particle an uncontrollable kick. Velocity can still be measured well, but only by very gentle long-wavelength position measurements spread over a long time, and therefore only at the cost of poor position information.

Only after these physical examples have dismantled the classical picture does Susskind make the mathematical pivot. Classical states are points in a set. Quantum states will live in a complex

vector space. Functions, column vectors, and complex numbers themselves are the first examples. That is where the lecture stops: not yet with the full machinery of operators and eigenvalues, but with the clear sense that some new logic is required, and with the first abstract language in which that logic can be expressed.

Chapter 2

Vector Spaces, Superposition, and the First Quantum Postulates

These notes follow Leonard Susskind's second quantum mechanics lecture as a guided argument rather than as a list of facts. The lecture begins by removing a conceptual obstruction left over from the discussion of the uncertainty principle, then turns to the formal language of complex vector spaces, bras and kets, inner products, bases, and finally the first quantum-mechanical postulates in the simple language of coins and dice. The selected lecture frames are kept where the board layout or the chalk geometry matters, but the transcript remains the primary source of order and emphasis. In that spirit we keep explicit credit to Leonard Susskind, together with the lecture-frame curation work of LazyingArt LLC and Video2Book.

2.1 Uncertainty Principle Prelude

The lecture does not begin with vector spaces. It begins by clearing away an objection. Suppose we measure the position of a particle only very poorly, to within a region of size δ, and then wait a very long time T before measuring its position again, again only to within a region of size δ. If the centers of the two fuzzy regions are separated by a distance L, then one seems to know the average velocity very accurately:

$$v_{\text{avg}} \sim \frac{L}{T}, \tag{2.1}$$

$$\Delta v_{\text{avg}} \sim \frac{\delta}{T}. \tag{2.2}$$

Since T may be made arbitrarily large, the uncertainty δ/T may be made arbitrarily small. At first sight this looks like a way of knowing position only to accuracy δ, while knowing velocity essentially exactly.

That, however, is not the quantity that the uncertainty principle is about. What we wanted was the position and momentum of the same state of the particle, or at least the momentum just before the position measurement. The two-measurement trick does not return that momentum.

2.1.1 Question & Answer

Can two imprecise position measurements defeat the uncertainty principle?

No. The first position measurement already disturbs the momentum. To localize a particle to scale δ, one must probe it with radiation of wavelength of order δ or smaller, and this produces a momentum kick of order

$$\Delta p \sim \frac{\hbar}{\delta}. \tag{2.3}$$

So the later motion reveals the velocity after the first measurement, not the original velocity of the undisturbed state. The uncertainty principle is not being evaded; rather, the wrong velocity is being inferred.

The lecture also presses the point of repeatability. One may say, "I think I measured the velocity," but if one tries to check that velocity one changes it again by an amount of order $\hbar/\delta$. In classical physics one imagines measuring position arbitrarily gently and leaving the velocity alone. In quantum mechanics, not so.

With that conceptual obstacle removed, the lecture pivots very explicitly: now we are ready to begin.

2.2 Complex Numbers as the Simplest Complex Vector Space

The next step is intentionally modest. Susskind says he is not trying to teach complex numbers; he wants only a very quick review, just for notational reasons. The subject of linear vector spaces may be dry, he says, but the good news is that it is also easy. So we begin with the easiest complex vector space of all: the complex numbers themselves.

A complex number is

$$z = x + iy, \qquad x, y \in \mathbb{R}, \tag{2.4}$$

and its complex conjugate is

$$z^* = x - iy. \tag{2.5}$$

Geometrically, z^* is the reflection of z across the real axis in the complex plane. The lecture briefly reminds us not to confuse that plane with the vector space we ultimately care about. The point of the example is structural: complex numbers can be added, and they can be multiplied by complex numbers, so they already form a one-dimensional complex vector space.

At this stage the bra-ket notation is introduced in embryo. In the one-dimensional case one may think of z as a ket and z^* as the corresponding bra. This is not yet deep; it is just a simple rehearsal for what will happen in more complicated vector spaces.

The lecture then isolates one algebraic fact that will return later in a more abstract form. If

$$z_1 = x_1 + iy_1, \qquad z_2 = x_2 + iy_2, \tag{2.6}$$

then

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2). \tag{2.7}$$

Complex conjugation flips only the sign of the imaginary part, and therefore

$$(z_1 z_2)^* = z_1^* z_2^*. \tag{2.8}$$

A related identity, written in the notation that foreshadows bras and kets, is

$$(z_1^* z_2)^* = z_2^* z_1. \tag{2.9}$$

For ordinary numbers the order is harmless, since the factors commute. But the lecture lingers on this order-swapping because it prepares the listener for the later statement that interchanging bra and ket corresponds to complex conjugation.

Only after this warm-up does the lecture state the abstract idea. A complex vector space is made of objects that can be added and multiplied by arbitrary complex numbers, and that stay within the space under those operations. In particular, if A and B are vectors and $\alpha, \beta \in \mathbb{C}$, then

$$\alpha A + \beta B \tag{2.10}$$

is again a vector of the same space.

That definition is deliberately spare. Vectors are not defined as arrows in physical space. They are defined by the operations we are allowed to perform on them. Ordinary arrows form only a real vector space, because multiplication by a general complex number is not defined for them. The real numbers form a real vector space, but not a complex one. Even the positive real numbers fail to form a real vector space, because multiplication by a negative number takes us out of the set. This operational point of view is the real beginning of the mathematics.

2.3 Complex Vector Spaces: Examples, Dimension, and Abstract Notation

Once the definition is on the table, the lecture immediately asks for examples. The first important example is the space of complex-valued functions of a real variable:

$$\psi(x) = \psi_{\mathrm{Re}}(x) + i\,\psi_{\mathrm{Im}}(x). \tag{2.11}$$

Here x is real, but the value of $\psi(x)$ is complex. If we multiply $\psi(x)$ by a complex number, we get another complex-valued function. If we add two such functions, we get another. In this mathematical sense, functions are vectors.

The second example is the column vector. A column vector is really a function of a discrete index. In the four-component case we may write

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}, \qquad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}, \tag{2.12}$$

and define the vector-space operations component by component:

$$|a\rangle + |b\rangle = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \\ a_4 + b_4 \end{pmatrix}, \tag{2.13}$$

$$\alpha|a\rangle = \begin{pmatrix} \alpha a_1 \\ \alpha a_2 \\ \alpha a_3 \\ \alpha a_4 \end{pmatrix}. \tag{2.14}$$

Thus column vectors are simply the discrete analog of complex functions. The lecture is careful here: if one is used to thinking of vectors only as arrows, one may not be used to thinking of functions as vectors. But that unfamiliarity is psychological, not mathematical.

Having supplied examples, the lecture turns to dimension. The dimension of a vector space is defined as the minimum number of vectors needed so that every vector in the space can be written as a linear combination of them. On a line, one nonzero vector suffices. On the blackboard plane, two non-collinear vectors suffice. This is the point at which the lecture draws the spanning picture.

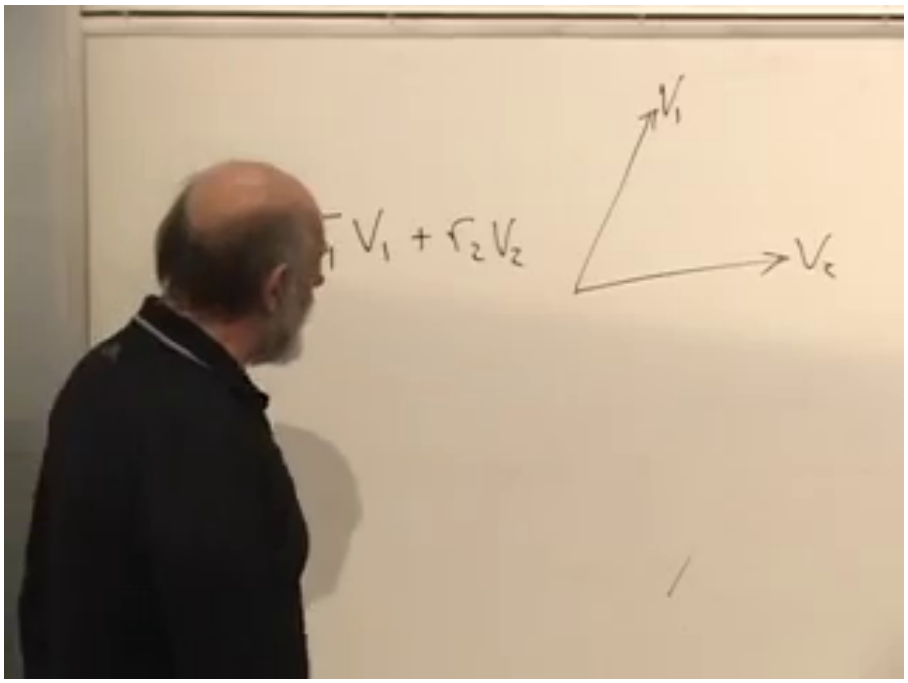


Figure 2.1: Linear combination from two spanning vectors. The chalk labels look closer to uppercase V_1, V_2, while the notes normalize the mathematics to lowercase v_1, v_2.

The chalk expression at the left of the board is partially obscured, but the clean mathematical statement is straightforward: if v_1 and v_2 are not on the same line, then every vector in the plane can be written as

$$r_1 v_1 + r_2 v_2. \tag{2.15}$$

A third vector v_3 drawn in the wedge between them would be redundant, because it can already be expressed in terms of the first two:

$$v_3 = r_1 v_1 + r_2 v_2. \tag{2.16}$$

The selected frame does not actually show v_3; that vector comes from the spoken argument and is drawn only in the cleaned reconstruction below.

From here the lecture runs through a sequence of dimensions. A four-component column space is four-dimensional. A one-component column space is just the one-dimensional complex vector space of ordinary complex numbers. Smooth functions of a real variable form an infinite-dimensional vector space, because no finite set of functions can be combined to generate every smooth function.

Only after that progression does the lecture shift notation. The point is to detach the reader from any one representation and move to the abstract vector itself.

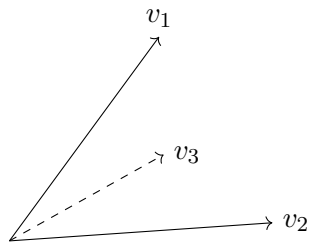


Figure 2.2: Clean reconstruction of the spanning argument. The dashed vector v_3 is reconstructed from the lecture discussion and is not visibly completed in the screenshot.

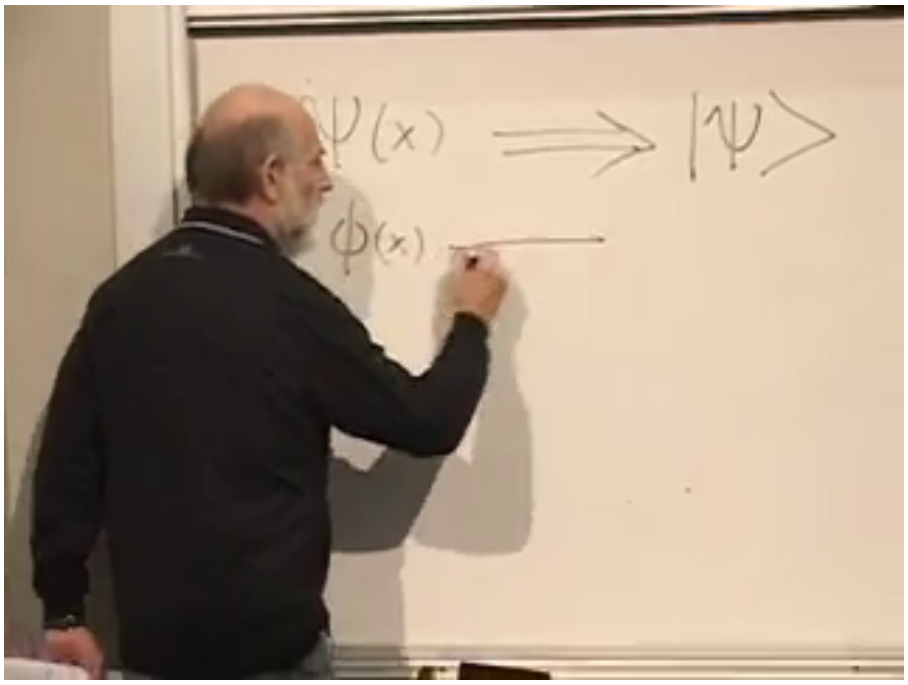


Figure 2.3: Mapping $\psi(x)$ to ket notation. The upper relation is fully visible on the board; the lower line is only partially visible and is completed cautiously in the text.

The board shows the completed line

$$\psi(x) \Rightarrow |\psi\rangle \tag{2.17}$$

and begins a second analogous line. In the text we complete that second line cautiously as

$$\phi(x) \mapsto |\phi\rangle. \tag{2.18}$$

The arrow here is not a strict equality sign. It is a change of viewpoint. We are now to think of the function as a vector in an abstract complex vector space. In the same way, a column of entries may be thought of as the abstract vector determined by those entries. This is the point at which the lecture's working notation settles into place.

2.4 Dual Vectors and Inner Products

Once abstract vectors are in place, the lecture introduces the dual vector space. For the purposes of this course, the dual space is the complex-conjugate partner of the original ket space. For every ket $|A\rangle$ there is a corresponding bra $\langle A|$. Mathematicians can define duals more generally, but the physical case under discussion is the complex-conjugate one.

In the case of functions, the bra corresponding to $|\psi\rangle$ is represented by the complex-conjugate function. In the case of column vectors, the corresponding bra is a row vector whose entries are the complex conjugates of the entries of the column. The lecture emphasizes that some of this is notation, but notation chosen to keep track of which objects live in the ket space and which in the bra space.

The first nontrivial rule concerns scalar multiplication. If $\langle A|$ corresponds to $|A\rangle$, then the bra associated with $\alpha|A\rangle$ is

$$\alpha^*\langle A|, \tag{2.19}$$

not $\alpha\langle A|$. This is just the familiar rule that complex conjugation of a product conjugates both factors. The lecture even pauses over a student's question about whether the α^* has to be written on the right; for ordinary numbers it does not matter, though later it will matter once operators are introduced.

The next construction is the inner product. Susskind introduces it by analogy with the ordinary dot product of real vectors. For ordinary vectors,

$$a \cdot b = a_x b_x + a_y b_y + a_z b_z + \cdots . \tag{2.20}$$

The general inner product $\langle A|B\rangle$ is the complex analogue of this idea, but now one must decide which vector is the bra and which is the ket. Once that choice is made, the inner product is required to satisfy the following properties:

$$\langle A|(\beta B)\rangle = \beta\langle A|B\rangle, \tag{2.21}$$

$$\langle A|(B+C)\rangle = \langle A|B\rangle + \langle A|C\rangle, \tag{2.22}$$

$$\langle A|B\rangle = \langle B|A\rangle^*, \tag{2.23}$$

$$\langle A|A\rangle > 0 \qquad \text{for } |A\rangle \neq 0. \tag{2.24}$$

From the third relation we learn immediately that $\langle A|A\rangle$ is real, since it equals its own complex conjugate. But the lecture insists on something subtler: positivity does *not* follow from the other properties. It is an additional part of the definition. Susskind even mentions relativity as an example where a bilinear form on a real vector space need not be positive.

The scalar model for positivity is the elementary identity

$$z^* z = (x - iy)(x + iy) = x^2 + y^2, \tag{2.25}$$

which is real and nonnegative. That is the pattern we carry into vector spaces.

The lecture then gives the concrete inner products we will actually use. For complex-valued functions of a real variable,

$$\langle\phi|\psi\rangle = \int dx\, \phi^*(x)\psi(x). \tag{2.26}$$

The lecture allows that functions of several real variables would simply require several integrations, but keeps the arguments real and the example simple.

For column vectors, if

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \qquad \langle b| = \begin{pmatrix} b_1^* & b_2^* & b_3^* \end{pmatrix}, \tag{2.27}$$

then

$$\langle b|a\rangle = b_1^* a_1 + b_2^* a_2 + b_3^* a_3 = \sum_i b_i^* a_i. \tag{2.28}$$

The lecture briefly corrects itself on the blackboard here, because one must remember that the row entries are the conjugates. In the final notes it is best simply to present the corrected form.

2.5 Orthonormal Bases and Expansion Coefficients

After the break the lecture becomes constructive again. We now introduce basis vectors, and again the order of ideas matters. First come unit length, then orthogonality, then orthonormality, and only then the practical use of the basis.

Let $\{|b_i\rangle\}_{i=1}^d$ be a collection of vectors in a d-dimensional vector space. They are normalized if

$$\langle b_i|b_i\rangle = 1 \qquad \text{for each } i, \tag{2.29}$$

with no sum on i. They are orthogonal if

$$\langle b_i|b_j\rangle = 0 \qquad \text{for } i \neq j. \tag{2.30}$$

Together these two conditions are summarized by

$$\langle b_i|b_j\rangle = \delta_{ij}, \tag{2.31}$$

where the Kronecker delta is defined by

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \tag{2.32}$$

A set of vectors satisfying this relation is orthonormal, and if there are d of them in a d-dimensional space, they form a basis.

The lecture illustrates this with the standard basis for a four-component column space. In cleaned notation we may write

$$|b_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |b_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |b_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |b_4\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \tag{2.33}$$

The corresponding bras are the conjugate row vectors. These vectors are normalized because each has only one nonzero entry, and they are orthogonal because two distinct basis vectors never overlap in the same slot. The lecture corrects a bra-versus-ket slip while discussing this example; here we simply present the corrected basis cleanly.

Now comes the practical payoff. Any vector $|v\rangle$ in a d-dimensional space can be expanded as

$$|v\rangle = \sum_{i=1}^{d} v_i |b_i\rangle. \tag{2.34}$$

The coefficients v_i are determined by taking inner products with the basis vectors. If we multiply on the left by $\langle b_j|$, we obtain

$$\langle b_j|v\rangle = \sum_{i=1}^{d} v_i \langle b_j|b_i\rangle \tag{2.35}$$

$$= \sum_{i=1}^{d} v_i \delta_{ji} \tag{2.36}$$

$$= v_j. \tag{2.37}$$

So the expansion coefficients are

$$v_j = \langle b_j|v\rangle. \tag{2.38}$$

This is the sequence the lecture wants us to see: first expand, then take the inner product, then use orthonormality, then discover the coefficients. Substituting back gives the compact formula that will recur throughout the course:

$$|v\rangle = \sum_{i=1}^{d} |b_i\rangle\langle b_i|v\rangle. \tag{2.39}$$

In the lecture the local blackboard derivation is a little messy; in the notes it is best presented in this cleaned form while preserving the same logical order.

2.6 Quantum States as Vectors: Coin, Die, Superposition, and Normalization

Only after the mathematics has been assembled does the lecture turn to physical meaning. The contrast with classical mechanics is drawn sharply. In classical mechanics, the space of states is a set. One speaks of unions and intersections, of AND and OR. In quantum mechanics, states are vectors, and the algebra is altogether different.

The lecture begins with the coin. Classically there are two states, heads and tails. One may imagine labeling them by symbols, but there would be no point in taking linear combinations of those labels. In quantum mechanics, by contrast, we assign to the two distinguishable configurations two orthogonal basis vectors, which we denote by $|H\rangle$ and $|T\rangle$. These are not lengths $+1$ and -1; they are simply names for two basis states. Their mathematical properties are

$$\langle H|T\rangle = 0, \qquad \langle H|H\rangle = \langle T|T\rangle = 1. \tag{2.40}$$

The orthogonality is not arbitrary. It is a postulate with physical meaning: states that can be distinguished by a single clean measurement are represented by orthogonal vectors. For the coin, the clean measurement is simply to look and ask whether the outcome is heads or tails. The same principle later applies to spin up and spin down, which is the real quantum-mechanical example lurking behind the coin.

Once the basis states are in place, the new quantum feature appears. The expression

$$|cc\rangle = \alpha_H|H\rangle + \alpha_T|T\rangle \tag{2.41}$$

is meaningful. Susskind calls this the state of a "confused coin." In classical physics the linear combination would mean nothing at all. In quantum mechanics it is a genuine state, different from both $|H\rangle$ and $|T\rangle$. The same idea applies to the die:

$$|cd\rangle = \sum_{i=1}^{6} \alpha_i|i\rangle, \tag{2.42}$$

where $|1\rangle, \ldots, |6\rangle$ are six orthonormal basis states corresponding to the six distinguishable outcomes of a die. Again, the labels $1, \ldots, 6$ are names, not vector lengths.

2.6.1 Question & Answer

How do the amplitudes become probabilities?

At this stage the lecture treats this as a postulate, not as something derived. If the confused coin is in the state

$$|cc\rangle = \alpha_H|H\rangle + \alpha_T|T\rangle, \tag{2.43}$$

then the probability of finding heads is

$$p_H = \alpha_H^*\alpha_H = |\alpha_H|^2, \tag{2.44}$$

and the probability of finding tails is

$$p_T = \alpha_T^*\alpha_T = |\alpha_T|^2. \tag{2.45}$$

This is the Born rule in its simplest discrete form.

The lecture is very careful here. This is not said to be the whole meaning of the amplitudes α_H and α_T. There is more in them than probabilities. But one feature of their meaning is already fixed: the squared moduli are the probabilities for measurement outcomes in the chosen orthonormal basis.

Operationally, these probabilities are understood from large ensembles of identically prepared systems. One prepares many copies of the same state $|cc\rangle$, performs the same measurement on each copy, and reads probabilities from the frequencies of outcomes. That is the level at which the postulate meets experiment.

Normalization now becomes unavoidable:

$$|\alpha_H|^2 + |\alpha_T|^2 = 1. \tag{2.46}$$

In bra-ket language this is simply

$$\langle cc|cc\rangle = 1. \tag{2.47}$$

Likewise, for the confused die,

$$\sum_{i=1}^{6} |\alpha_i|^2 = 1, \qquad \langle cd|cd\rangle = 1. \tag{2.48}$$

Thus the lecture arrives at another postulate: physically admissible states are normalized.

A useful explicit example is the equal-probability state. If the confused coin gives heads and tails with equal probability, then

$$|\alpha_H| = |\alpha_T| = \frac{1}{\sqrt{2}}. \tag{2.49}$$

But the amplitudes are complex, so the general equal-probability form is

$$\alpha_H = \frac{e^{i\theta_H}}{\sqrt{2}}, \qquad \alpha_T = \frac{e^{i\theta_T}}{\sqrt{2}}. \tag{2.50}$$

Similarly, an equal-probability die state may be written as

$$\alpha_i = \frac{e^{i\theta_i}}{\sqrt{6}}. \tag{2.51}$$

The lecture stresses exactly this point: equal probability does not mean the amplitudes are literally $1/\sqrt{2}$ or $1/\sqrt{6}$; it means they have those moduli, up to unit complex phases.

By the end of this section the first postulates have been stated in the language of simple discrete systems:

1. Clearly distinguishable configurations correspond to orthogonal vectors.
2. Physical states are normalized.
3. In a basis of distinguishable states, probabilities are given by squared moduli of amplitudes.

The deeper meaning of superposition is not yet exhausted, but the lecture has now established its first operational consequences.

2.7 Operators as the Next Mathematical Step

There is one more turn before the lecture ends. Susskind says he would like to get to operators, and he does, but only as an opening move. The subject is introduced first through ordinary blackboard vectors, before any full physical interpretation is attached to it.

The sketch is intentionally elementary: a vertical guide and two ordinary vectors. The question is what kinds of operations one can do on all vectors in such a space. One may rotate every vector by the same angle, double every vector while keeping its direction fixed, or reflect every vector about an axis. These are genuine operations on the vector space.

Not every operation is linear. The lecture gives a deliberately simple counterexample: replace every vector of length 2 by one of length 4, every vector of length 3 by one of length 9, and so on. That is certainly an operation, but it is not linear.

A linear operator, denoted with a hat, is characterized by two properties:

$$\hat{L}(\alpha a) = \alpha\, \hat{L}(a), \tag{2.52}$$

$$\hat{L}(a + b) = \hat{L}(a) + \hat{L}(b). \tag{2.53}$$

The lecture checks these properties on the simplest geometric example. If R denotes rotation by a fixed angle, then

$$R(2a) = 2R(a), \tag{2.54}$$

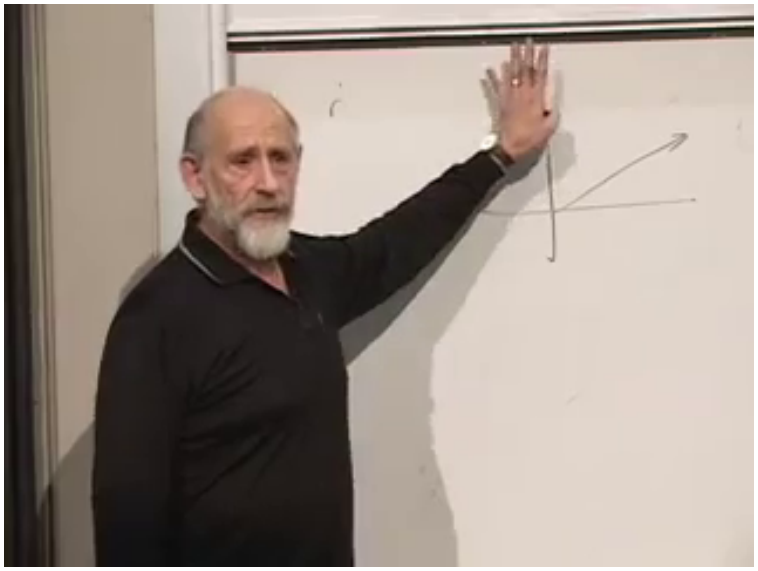


Figure 2.4: Ordinary vectors drawn on the blackboard. The figure is diagrammatic rather than symbolic, and serves as evidence for the geometric setup used to introduce operators.

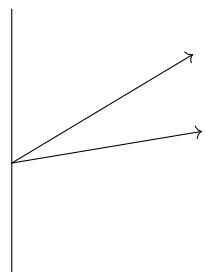


Figure 2.5: Clean redraw of the ordinary-vector sketch used in the lecture's first discussion of operations on vectors.

and if $a+b$ is drawn as the diagonal of the usual parallelogram, then rotating the whole parallelogram shows that

$$R(a+b) = R(a) + R(b). \tag{2.55}$$

So fixed-angle rotation is linear. Uniform doubling is linear. Reflection is linear. Squaring the length is not.

This final section is explicitly transitional. The lecture reaches just far enough to state the physical slogan: states are represented by vectors, but measurable quantities are represented by linear operators. In that sense observables such as the "headsness" or "tailsness" of the coin belong on the operator side of the formalism. The actual identification of observables with operators is postponed to the next lecture.

2.8 Summary

This lecture unfolds in two layers. First it clears away a conceptual misunderstanding about the uncertainty principle. Then it builds the mathematical language in which the first quantum

postulates can be stated.

We began by seeing why two poor position measurements far apart in time do not reveal the original momentum of a particle: the first position measurement itself changes the momentum by an amount of order $\hbar/\delta$. From there the lecture reset and moved through complex numbers, complex vector spaces, concrete examples in functions and column vectors, the notion of dimension, the shift to abstract kets, the dual space, inner products, orthonormal bases, and the extraction of expansion coefficients.

Only after that apparatus was in place did the lecture turn to physics. Classical states form a set; quantum states form a vector space. Distinguishable configurations correspond to orthogonal vectors, superpositions are new states, normalization encodes total probability one, and squared amplitudes give measurement probabilities in a basis of distinguishable states. Finally, the lecture opened the next mathematical subject, operators, just far enough to show the direction of travel. The formalism is still abstract here, but the chapter ends with the same forward pressure as the lecture itself: we are now very close to the point where these rules begin to make concrete predictions.

Chapter 3

Linear Operators, Hermitian Observables, and a Particle on a Line

We begin where the lecture begins: not with a new subject, but with a review. The mathematics of quantum mechanics is linear algebra, and the first order of business is to keep two ideas distinct. States are vectors. What we measure are not states, but observables, and observables are represented by Hermitian linear operators acting on those vectors. The lecture then unfolds in a precise order: first the language of operators and bases, then the passage to matrices, then Hermitian operators and eigenvectors, then the postulates, and only after that the first truly infinite-dimensional example, a particle moving on a line. A few board frames are preserved here as visual evidence, with credit to Leonard Susskind, LazyingArt LLC, and Video2Book.

3.1 Linear Operators, Matrix Elements, and Bases

The opening insistence is worth preserving: quantum mechanics is linear algebra. We start with a vector space, because states will live in that space, and we introduce linear operators, because measurable quantities will eventually be represented by a special class of them.

A linear operator is denoted by a capital letter with a hat. It acts on a ket and produces another ket:

$$\hat{K}|A\rangle = |C\rangle. \tag{3.1}$$

This much is still abstract, but the inner product immediately allows us to turn the action of an operator into an ordinary number. If we let $\hat{K}$ act on $|A\rangle$, and then take the inner product with $\langle B|$, we obtain the matrix element

$$\langle B|\hat{K}|A\rangle. \tag{3.2}$$

The notation is meant to be read in that order: first $\hat{K}$ acts on $|A\rangle$, then we take the inner product with $\langle B|$.

Now choose an orthonormal basis $\{|n\rangle\}$. If the vector space has dimension D, then the basis contains D independent vectors, mutually orthogonal and of unit norm. The lecture pauses here to draw an ordinary three-dimensional picture, not because the quantum vector space is literally $\mathbb{R}^3$, but because complex vector spaces are hard to draw and one still wants a geometric picture of what a basis is. There can be many different bases, tilted relative to one another, and that visual fact will later become physically important.

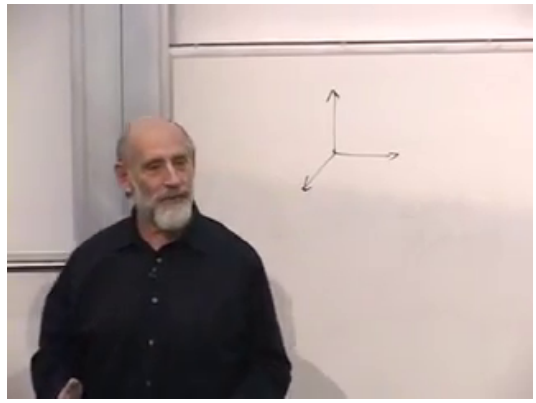


Figure 3.1: Ordinary three-vector basis sketch on the board.

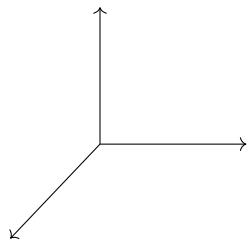


Figure 3.2: Transcript-based redraw of the basis sketch. The point is not the literal geometry, but the existence of distinct orthogonal basis directions.

Once a basis has been chosen, we can tabulate the operator by its matrix elements between basis vectors. The board isolates this notation very clearly:

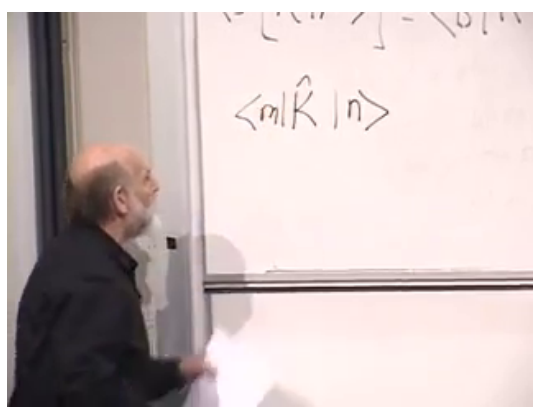


Figure 3.3: Matrix element between basis vectors.

$$\langle m|\hat{K}|n\rangle. \tag{3.3}$$

If we define

$$K_{mn} := \langle m|\hat{K}|n\rangle, \tag{3.4}$$

then the operator is represented, in that basis, by a square matrix of numbers. The lecture emphasizes the simple counting: if there are D basis vectors, there are D^2 such matrix elements. Once the basis is fixed, these entries describe the operator completely.

3.2 From Abstract Kets to Matrix Multiplication

The next step is to turn vectors themselves into components. If $\{|n\rangle\}$ is a basis, any vector $|A\rangle$ can be expanded as

$$|A\rangle = \sum_n A_n|n\rangle, \qquad A_n = \langle n|A\rangle. \tag{3.5}$$

So the coefficients are not mysterious new objects; they are just inner products with basis vectors.

Substituting the second formula into the first gives

$$|A\rangle = \sum_n |n\rangle\langle n|A\rangle. \tag{3.6}$$

The lecture describes the operator in the middle as something that "does nothing." In compact notation we write

$$\sum_n |n\rangle\langle n| = \mathbb{I}. \tag{3.7}$$

This is the completeness relation. The lecture itself does not pause to write $\mathbb{I}$, but that is the clean modern way to say what is meant by those basis vectors "nose to nose" under a summation sign.

Now we go back to the operator $\hat{K}$ acting on an arbitrary vector $|A\rangle$. The goal is not merely to write $\hat{K}|A\rangle$ abstractly, but to compute its components. The n-th component is

$$\langle n|\hat{K}|A\rangle. \tag{3.8}$$

Insert the expansion of $|A\rangle$:

$$\langle n|\hat{K}|A\rangle = \left\langle n \middle| \hat{K}\sum_m A_m|m\rangle \right\rangle \tag{3.9}$$

$$= \sum_m \langle n|\hat{K}|m\rangle A_m \tag{3.10}$$

$$= \sum_m K_{nm}A_m. \tag{3.11}$$

This is the exact point at which the lecture moves from abstract bra-ket notation to ordinary matrix language.

The upper line on the board is the bra-ket statement

$$\langle n|\hat{K}|A\rangle = \sum_m \langle n|\hat{K}|m\rangle\langle m|A\rangle, \tag{3.12}$$

and the lower line compresses the same content into the indexed form

$$(\hat{K}A)_n = \sum_m K_{nm}A_m. \tag{3.13}$$

That is just matrix multiplication: the matrix K_{nm} acts on the column of components A_m.

The lecture then pushes this one step further. If two operators $\hat{K}$ and $\hat{L}$ are multiplied, their product means that $\hat{L}$ acts first and $\hat{K}$ acts second. Since operators are not numbers, one must not assume that $\hat{K}\hat{L} = \hat{L}\hat{K}$. In general they do not commute. The matrix elements of the product are

$$(KL)_{nm} = \langle n|\hat{K}\hat{L}|m\rangle \tag{3.14}$$

$$= \sum_r \langle n|\hat{K}|r\rangle\langle r|\hat{L}|m\rangle \tag{3.15}$$

$$= \sum_r K_{nr}L_{rm}. \tag{3.16}$$

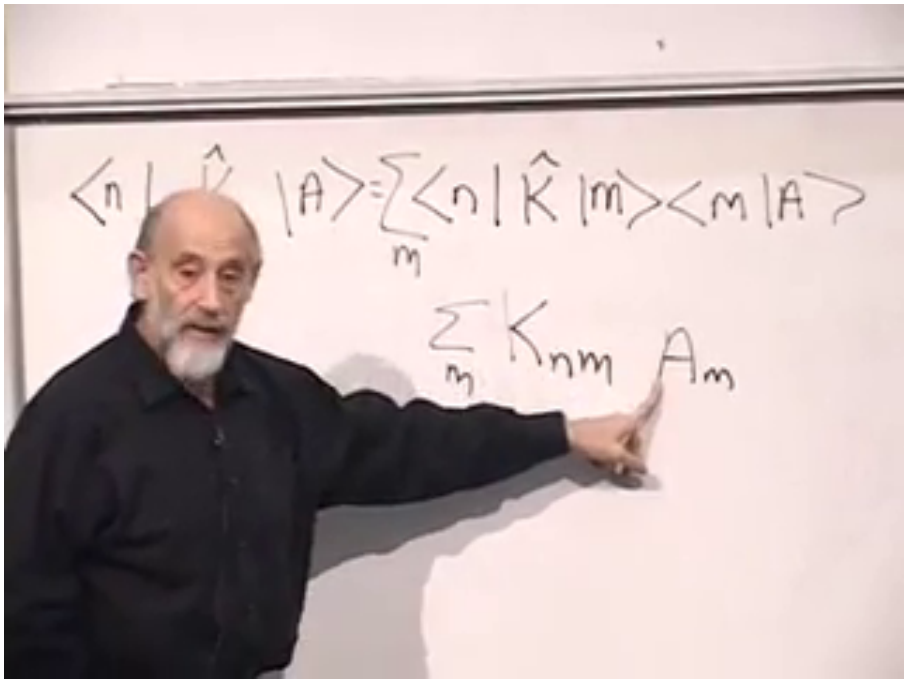


Figure 3.4: Operator action rewritten in components.

So the abstract rule "do one thing and then another" becomes the concrete rule of matrix multiplication. This is the last piece of the linear-algebra toolkit the lecture wants in place before turning to the operators of physical interest.

3.3 Hermitian Operators and the Eigenvalue Theorem

The next idea is the decisive one. Hermitian operators play, in operator theory, the role that real numbers play among ordinary numbers. They are the operators appropriate to measurable quantities, because measurements return real values.

The first chalkboard statement of the Hermitian definition is garbled in the transcript, but the later repetitions make the intended rule clear. We therefore state the standard form.

Definition 3.1. An operator $\hat{H}$ is Hermitian if

$$\langle B|\hat{H}|A\rangle = \langle A|\hat{H}|B\rangle^*. \tag{3.17}$$

Equivalently, for every vector $|A\rangle$,

$$\langle A|\hat{H}|A\rangle \in \mathbb{R}. \tag{3.18}$$

This is the precise sense in which Hermitian operators are "real." If we lived in a purely real vector space, the analogue would simply be symmetry: the matrix elements would satisfy $H_{mn} = H_{nm}$.

To appreciate the significance of Hermitian operators, the lecture turns immediately to eigenvectors and eigenvalues. In general an operator changes the direction of a vector. But there may be special directions that it leaves unchanged. If $|\lambda\rangle$ is such a vector, then

$$\hat{H}|\lambda\rangle = \lambda|\lambda\rangle. \tag{3.19}$$

The vector is an eigenvector; the number λ is the corresponding eigenvalue.

Theorem 3.2 (Fundamental facts about Hermitian operators)**.** *If $\hat{H}$ is Hermitian, then:*

1. *every eigenvalue of $\hat{H}$ is real;*
2. *eigenvectors corresponding to distinct eigenvalues are orthogonal;*
3. *the eigenvectors of $\hat{H}$ form a complete orthonormal basis of the vector space. In the degenerate case, one can choose orthogonal eigenvectors within each degenerate eigenspace.*

The lecture does not stop to prove the full theorem, but it does emphasize that the first two facts follow very quickly from the definition. Let us record them.

Proof. Suppose

$$\hat{H}|\lambda\rangle = \lambda|\lambda\rangle. \tag{3.20}$$

Then

$$\langle\lambda|\hat{H}|\lambda\rangle = \lambda\langle\lambda|\lambda\rangle. \tag{3.21}$$

The left-hand side is real, because $\hat{H}$ is Hermitian. The factor $\langle\lambda|\lambda\rangle$ is positive for a nonzero vector. Hence λ must be real.

Now suppose

$$\hat{H}|\lambda_1\rangle = \lambda_1|\lambda_1\rangle, \qquad \hat{H}|\lambda_2\rangle = \lambda_2|\lambda_2\rangle. \tag{3.22}$$

Then

$$\langle\lambda_1|\hat{H}|\lambda_2\rangle = \lambda_2\langle\lambda_1|\lambda_2\rangle. \tag{3.23}$$

On the other hand, Hermiticity gives

$$\langle\lambda_1|\hat{H}|\lambda_2\rangle = \langle\hat{H}\lambda_1|\lambda_2\rangle = \lambda_1\langle\lambda_1|\lambda_2\rangle, \tag{3.24}$$

where we used the reality of λ_1. Therefore

$$(\lambda_2 - \lambda_1)\langle\lambda_1|\lambda_2\rangle = 0. \tag{3.25}$$

So if $\lambda_1 \neq \lambda_2$, then $\langle\lambda_1|\lambda_2\rangle = 0$. □

The third statement is the lecture's real payoff. Hermitian operators do not merely have a few privileged vectors; they have enough of them to form a basis. And because an eigenvector may be multiplied by any nonzero complex number without ceasing to be an eigenvector, we can always rescale it to unit norm. That is how the orthogonal eigenvectors become an orthonormal basis.

The lecture then returns to the concrete matrix picture. The board frame at this point shows the matrix and column-vector layout explicitly, though only the upper-left corner of the matrix is fully legible.

Remark 3.3. The frame shows the matrix entries $H_{11}, H_{12}, H_{21}, \ldots$ and the first components $A_1, A_2, \ldots$ of the column vector. The fully typeset array below is a cautious completion of that visible pattern, preserving the lecture's local use of H_{ij} rather than forcing everything back to K_{ij}.

$$\begin{pmatrix} H_{11} & H_{12} & \cdots \\ H_{21} & H_{22} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \\ \vdots \end{pmatrix}. \tag{3.26}$$

If the column happens to be an eigenvector of the matrix, then the result is simply λ times the same column vector. That is the concrete matrix form of the eigenvalue equation.

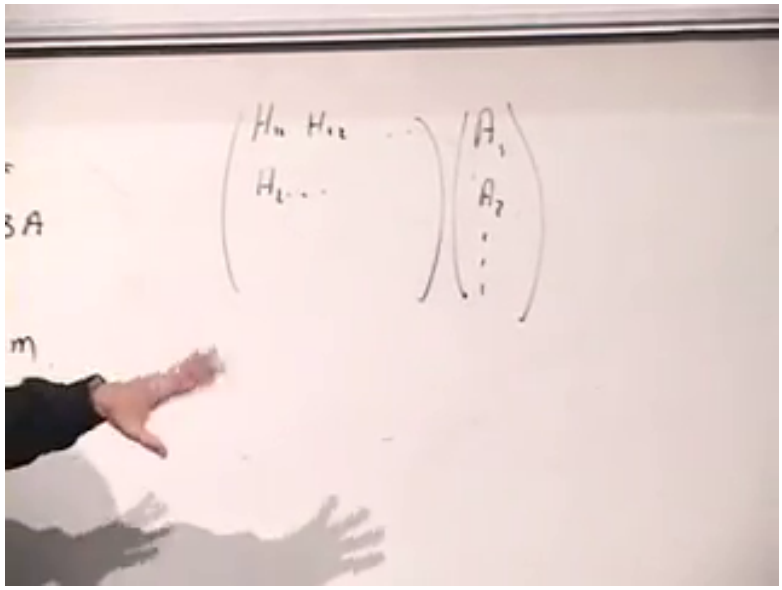


Figure 3.5: Concrete matrix-and-column-vector layout in the Hermitian-operator discussion.

3.3.1 Question & Answer

Do all operators have eigenvectors? No. The lecture makes a point of pausing here because this is exactly where one might overgeneralize. A generic linear operator need not have eigenvectors in the useful sense, and it certainly need not have a complete basis of them. Hermitian operators are special. They do.

This is also why the exchange about rotations matters. A rotation is not Hermitian. In a real vector space the corresponding matrices are antisymmetric, and in the complex setting one is led naturally to anti-Hermitian generators. So the theorem is not a fact about arbitrary operators. It is a fact about the Hermitian ones.

3.4 The Five Postulates of Quantum Mechanics

Only now does the lecture say that we are ready for the postulates of quantum mechanics. They are not presented as the smallest logically independent set. They are presented in a slightly expanded form because, pedagogically, it is better to expose the ideas than to be clever.

1. **States are vectors.** Every state of a quantum system is represented by a ket vector in a complex vector space.
2. **Observables correspond to Hermitian operators.** The things we measure in experiments are represented by Hermitian operators acting on the state space.
3. **Possible measurement outcomes are eigenvalues.** If an observable corresponds to $\hat{H}$, then the only values that can be obtained in measurement are the eigenvalues of $\hat{H}$.
4. **Eigenvectors are states of definite value.** If the system is in an eigenvector $|\lambda_n\rangle$ of $\hat{H}$, then a measurement of the observable yields the value λ_n with certainty.

5. **Arbitrary states give probabilities.** If the system is in an arbitrary state $|A\rangle$, then the probability of measuring λ_n is
$$P(\lambda_n|A) = |\langle\lambda_n|A\rangle|^2. \tag{3.27}$$

The lecture keeps correcting the language here, and those corrections are part of the substance. One does not directly observe the state of a system. One observes observables and records real numbers. The state is what determines the probabilities for those numbers.

3.4.1 Question & Answer

What does it mean for an observable to be *definite*, and is it right to say that we observe eigenvectors? In the lecture, a quantity is definite when there is no statistical spread in the outcome. If the system has been prepared in an eigenvector of $\hat{H}$, then repeated measurements of H always yield the same eigenvalue. The observable is definite in that state.

But it is still better not to say that we observe eigenvectors. The cleaner statement is that we measure an observable and obtain one of its eigenvalues. From that result we may infer something about the state, but the thing recorded by the measuring apparatus is a number.

The lecture illustrates this with spin. Prepare an electron in a spin eigenstate along one axis and measure spin along that same axis: the result is definite. Measure along a different axis: now the result is only probabilistic. The state has not ceased to exist; rather, it is not an eigenstate of the second observable.

Remark 3.4. At this stage the lecture has stated the probability rule, but it has not yet given an account of the measurement process itself. The effect of measurement on the state, and the issue of collapse, are explicitly postponed.

3.5 A Particle on a Line and the Position Basis

The lecture now makes a deliberate jump. Instead of returning to coins, dice, or other finite-state systems, it goes directly to a particle moving on a line. This is where the state space becomes genuinely infinite-dimensional.

Classically, a particle on a line is described by a coordinate x, perhaps depending on time. Quantum mechanically, the state is described instead by a vector in the space of complex functions of x. We write that vector as $\psi(x)$. This is not a metaphor. The set of such functions forms a vector space: we can add functions, multiply them by complex numbers, and define an inner product

$$\langle\phi|\psi\rangle = \int dx\, \phi^*(x)\psi(x). \tag{3.28}$$

The lecture also pauses here to head off a possible confusion. This inner product is not real in general. What is guaranteed real is the inner product of a vector with itself:

$$\langle\psi|\psi\rangle = \int dx\, |\psi(x)|^2. \tag{3.29}$$

Now we introduce the first operator in this function space: the position operator $\hat{x}$. Its action is as simple as possible:

$$(\hat{x}\psi)(x) = x\psi(x). \tag{3.30}$$

The lecture checks Hermiticity by using the equivalent condition that $\langle\psi|\hat{x}|\psi\rangle$ be real for every ψ:

$$\langle\psi|\hat{x}|\psi\rangle = \int dx\, \psi^*(x)\, x\, \psi(x) = \int dx\, x\, |\psi(x)|^2. \tag{3.31}$$

Because x and dx are real, this quantity is real. So $\hat{x}$ is Hermitian.

That means it must have eigenvectors. The eigenvalue equation is

$$\hat{x}|\lambda\rangle = \lambda|\lambda\rangle, \tag{3.32}$$

which in the function representation becomes

$$x\, \psi(x) = \lambda\psi(x), \tag{3.33}$$

or

$$(x - \lambda)\psi(x) = 0. \tag{3.34}$$

Taken literally, this looks absurd. For every $x \neq \lambda$, the factor $x - \lambda$ is nonzero, so $\psi(x)$ must vanish. The only place where it can fail to vanish is at one point. As an ordinary square-integrable function, that is impossible.

The lecture does not hide the difficulty. Instead it introduces the Dirac delta function heuristically. One imagines a narrow spike of width ϵ centered at $x = \lambda$, of height $1/\epsilon$, so that the area under it is one. In standard notation, the cleaned-up form of the lecture's spike picture is

$$\langle x|\lambda\rangle = \delta(x - \lambda). \tag{3.35}$$

This is not an ordinary normalized wavefunction. It is an idealized generalized eigenfunction.

The lecture also insists on another distinction that is easy to blur: these position eigenfunctions are orthogonal, but not normalized in the usual sense. If $\lambda \neq \lambda'$, then the corresponding spikes do not overlap, so their inner product vanishes. But the inner product of such a spike with itself is not a nice finite unit norm. In the lecture's language, it is "ortho, but not normal."

There is one such eigenfunction for every real value of λ. The spectrum of $\hat{x}$ is therefore the entire real axis.

The real payoff comes when we ask what the wavefunction $\psi(x)$ means. Project the state onto the position eigenvector at λ:

$$\langle\lambda|\psi\rangle = \int dx\, \delta(x - \lambda)\psi(x). \tag{3.36}$$

Now use the spike picture. Over the narrow support of the delta function, $\psi(x)$ varies negligibly, so we may replace it by $\psi(\lambda)$:

$$\langle\lambda|\psi\rangle \approx \psi(\lambda) \int dx\, \delta(x - \lambda) \tag{3.37}$$

$$= \psi(\lambda). \tag{3.38}$$

In the standard distributional reading, this is the familiar identity. Relabel the eigenvalue λ by the position variable itself, and we obtain

$$\langle x|\psi\rangle = \psi(x). \tag{3.39}$$

So the wavefunction is the component of the state vector along the position basis. It is not an extra object attached to the ket. It is the ket, expressed in a particular basis.

Now the fifth postulate becomes concrete. The probability is the square of the projection onto the relevant eigenvector. In the position basis that means

$$\rho(x) = |\psi(x)|^2. \tag{3.40}$$

The lecture phrases this informally as the probability to detect the particle at position x. Mathematically, for a continuous spectrum, it is better to say that $|\psi(x)|^2$ is the position probability density.

3.5.1 Question & Answer

How can position eigenvectors be represented if normalizable wavefunctions must decay at infinity? The lecture raises this difficulty directly and then refuses to overtidy it. Exact position eigenvectors are not ordinary normalizable Hilbert-space vectors. They are idealized objects, represented heuristically by delta spikes.

What one can do physically is approximate them arbitrarily well by sharply localized wave packets. That is enough for the lecture's immediate purpose. The normalization issue is real, but it is not the issue the lecture wants to settle at this moment.

3.6 The k-Operator, Plane Waves, and the Momentum Hint

At this point the lecture asks a very natural question. If all we care about is the position probability density $|\psi(x)|^2$, why go through the trouble of introducing a complex wavefunction $\psi(x)$ at all? Why not simply use a probability density on the line?

The answer is that position is not the only observable. We need a single state vector from which the probabilities for many different observables can be extracted. That is the reason the full wavefunction is needed.

The next operator is differentiation:

$$\frac{d}{dx}. \tag{3.41}$$

It is linear, but it is not Hermitian. The lecture's first blackboard definition of anti-Hermitian is also garbled, so here again we state the corrected standard form:

$$\langle B|\hat{A}|A\rangle = -\langle A|\hat{A}|B\rangle^*. \tag{3.42}$$

Under this rule, multiplying an anti-Hermitian operator by i produces a Hermitian one.

For differentiation, the cleaned-up operator statement is

$$\left\langle \phi \left| \frac{d}{dx} \right| \psi \right\rangle = -\left\langle \psi \left| \frac{d}{dx} \right| \phi \right\rangle^*, \tag{3.43}$$

provided the boundary terms vanish at infinity. The lecture then settles on the Hermitian combination

$$\hat{K} = -i\,\frac{d}{dx}. \tag{3.44}$$

There is a brief sign wobble in the live discussion, but $-i\,d/dx$ is where the lecture comes to rest, and that is the convention we keep.

The lecture chooses the equivalent Hermiticity check $\langle\psi|\hat{K}|\psi\rangle \in \mathbb{R}$. Let us follow that calculation.

Proof. We compute

$$\langle\psi|\hat{K}|\psi\rangle = \int dx\, \psi^*(x) \left(-i\, \frac{d\psi}{dx}\right). \tag{3.45}$$

Integrating by parts and dropping the boundary term—for the moment assuming that $\psi(x) \to 0$ sufficiently fast at infinity—gives

$$\int dx\, \psi^*(x) \left(-i\, \frac{d\psi}{dx}\right) = i \int dx\, \frac{d\psi^*}{dx}\, \psi(x). \tag{3.46}$$

But the right-hand side is the complex conjugate of the left-hand side. Therefore the original expression is equal to its own complex conjugate and is real. Hence $\hat{K}$ is Hermitian. □

The lecture is already aware of the tension. This proof assumes decay at infinity, but the eigenfunctions we are about to write down do not decay. That is not a mistake to be hidden. It is one of the reasons the discussion remains heuristic at this stage.

Now solve the eigenvalue equation

$$-i\, \frac{d\psi}{dx} = k\psi. \tag{3.47}$$

The solution is

$$\psi_k(x) \propto e^{ikx} = \cos(kx) + i \sin(kx). \tag{3.48}$$

The lecture does not belabor the solution method; it simply checks it:

$$\frac{d}{dx} e^{ikx} = ik\, e^{ikx}, \tag{3.49}$$

so multiplying by $-i$ yields $k\, e^{ikx}$, exactly as required.

These eigenfunctions are radically different from those of position. The position eigenfunctions are sharply peaked. The k-eigenfunctions are spread across the whole line. Moreover,

$$\psi_k^*(x)\psi_k(x) = \text{const.}, \tag{3.50}$$

so the corresponding position probability density is uniform. There is no information here about where the particle is.

The lecture then lingers on the oscillatory character of e^{ikx}. As x changes, the complex number e^{ikx} moves around the unit circle. Small k means slow winding, hence long wavelength. Large k means rapid winding, hence short wavelength. One full period occurs when kx changes by 2π, so

$$kL = 2\pi, \qquad L = \frac{2\pi}{k}. \tag{3.51}$$

Only after this mathematical work does the lecture attach a physical name. It does so cautiously. Using the pre-quantum intuition of de Broglie and Einstein, one recalls that a wave of wavelength L is associated with momentum

$$p = \frac{h}{L}. \tag{3.52}$$

Substituting $L = 2\pi/k$ gives

$$p = \frac{h}{L} = \frac{h}{2\pi} k = \hbar k. \tag{3.53}$$

So the operator $\hat{K}$ is provisionally interpreted as momentum in units of $\hbar$.

The lecture is explicit about the status of this statement. It is a name, not yet a derivation from first principles. To show that $\hbar k$ really behaves like classical momentum, one must study time evolution, wave packets, and the classical limit. That is postponed.

Even before that deeper analysis, however, the contrast with position is already stark. The position eigenbasis consists of sharply localized states. The k-basis consists of delocalized oscillatory states. No position eigenfunction is also a k-eigenfunction, and no k-eigenfunction is also a position eigenfunction. The lecture rightly hears, in that clash, the first unmistakable note of the uncertainty principle.

3.7 Summary

The lecture begins by rebuilding the machinery: operators act on kets, matrix elements turn operator action into numbers, and a basis turns vectors into components. That bridge to components is what lets linear operators become matrices and operator products become matrix multiplication.

From there the lecture narrows to the physically significant class of Hermitian operators, introduces eigenvalues and eigenvectors, and states the fundamental theorem that Hermitian operators have complete orthonormal eigenbases. Only then does it state the postulates of quantum mechanics and move to the first serious continuous example, a particle on a line.

In that example we learn two things at once. First, $\psi(x)$ is the component of the state along the position basis, and $|\psi(x)|^2$ is the position probability density. Second, there are other observables, represented by different Hermitian operators. The operator $\hat{K} = -i\, d/dx$ leads to plane-wave eigenfunctions e^{ikx}, to the wavelength relation $L = 2\pi/k$, and, provisionally, to the momentum identification $p = \hbar k$. What remains for later is exactly what the lecture says remains for later: the probability distribution for momentum, the role of measurement in changing the state, and the time-dependent route by which these quantum constructions recover the classical notion of momentum.

Chapter 4

From Discrete Observables to Interference

This chapter follows Leonard Susskind's fourth lecture in the quantum mechanics sequence and keeps its order and emphasis: review first, replacement second, technical detour third, then the return to probabilities, then a new system, then the question of compatibility, and finally the two-slit experiment. The lecture itself is Leonard Susskind's; these notes also preserve selected blackboard evidence, with curation support acknowledged to LazyingArt LLC and Video2Book. The governing idea is that the postulates of quantum mechanics do not change when the observable becomes continuous, but the mathematical objects that express those postulates do.

4.1 Review: Discrete Observables, Probabilities, and Basis States

Let us begin exactly where the lecture begins: with a measurable quantity that can take discrete values. Call those values $n = 1, 2, 3, \ldots$. If the system is in some state $|\psi\rangle$, then a measurement returns one of these values with probability $P(n)$, and the first rule is simply

$$\sum_n P(n) = 1. \tag{4.1}$$

The second rule is that the basis vectors associated with the different values of the observable may be taken to form an orthonormal basis:

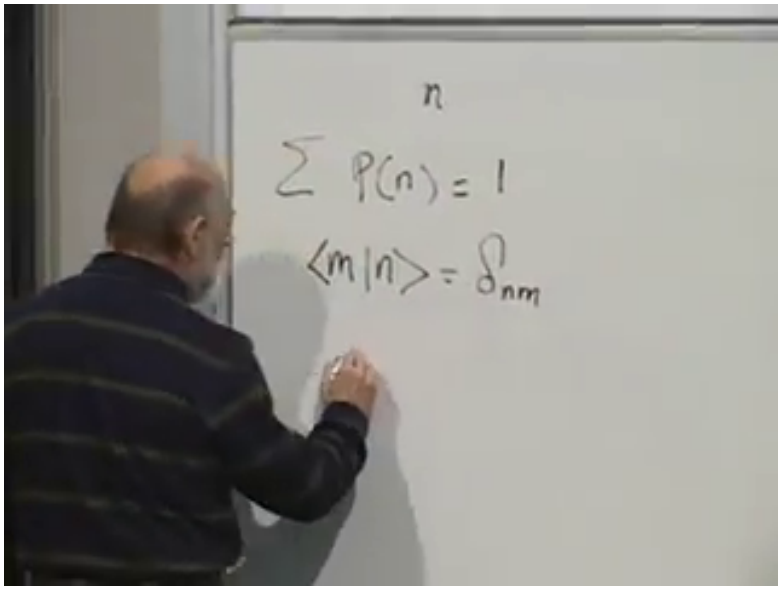
$$\langle m|n\rangle = \delta_{nm}. \tag{4.2}$$

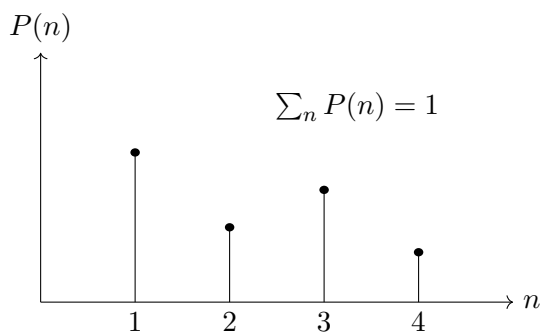
Here δ_{nm} is the Kronecker delta, equal to 1 when $m = n$ and 0 otherwise. Susskind pauses to note that one may think of it as depending only on the difference $n - m$; it is really the vanishing or nonvanishing of that difference that matters.

Probabilities are governed by amplitudes. In the discrete case the amplitude for obtaining the value n is $\langle n|\psi\rangle$, and the probability is the squared magnitude of that amplitude:

$$P(n) = |\langle n|\psi\rangle|^2. \tag{4.3}$$

So already, before any continuum enters, we have the full pattern: a basis of states, an inner product, amplitudes, and the rule that probabilities are obtained by squaring amplitudes.





The blackboard here still has the character of a review. The normalization and orthonormality relations are written cleanly at the top, while the probability picture is only beginning below. That is faithful to the lecture's rhythm: the picture is secondary; the formal rules are being re-established first.

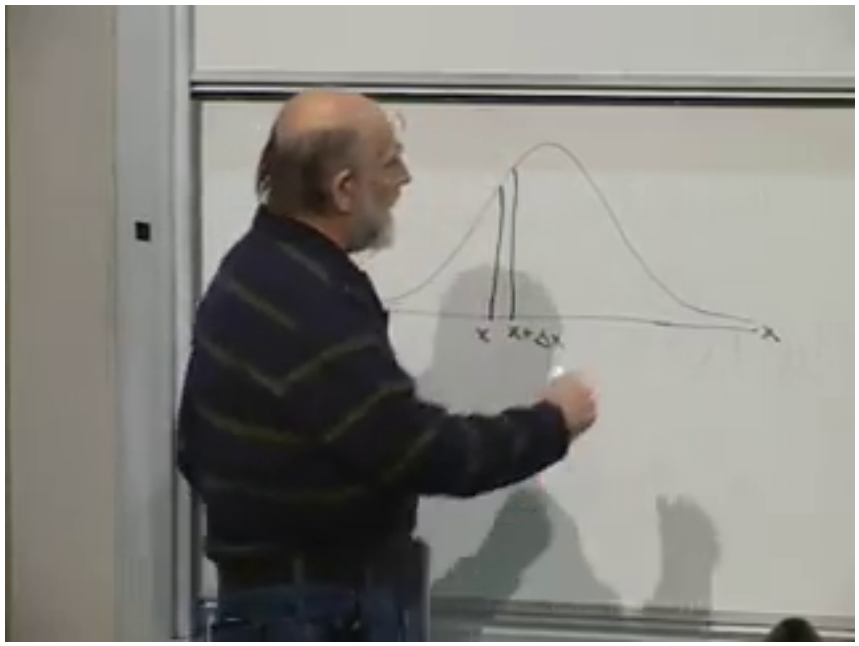
If we plot the probability as a function of the discrete variable n, the values live only on the integers. The picture is therefore not a continuous curve but a collection of heights, and it is those heights that add up to 1.

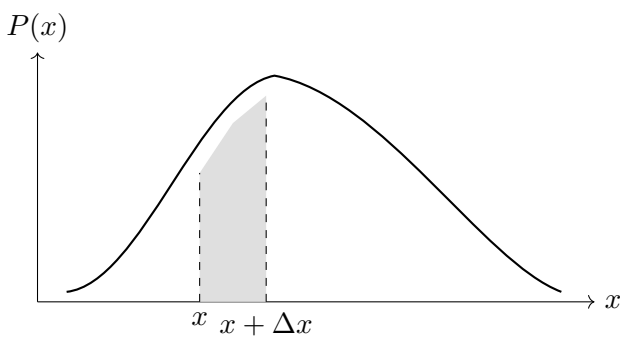
This review is not detachable background. It is the template for everything that follows. Once the observable becomes continuous, sums will be replaced by integrals and Kronecker deltas by Dirac deltas, but the structure of the argument remains the same.

4.2 From Discrete Probabilities to Continuous Probability Density

Now let the observable be the position of a particle on a line. The first change is conceptual rather than algebraic. It no longer makes sense to ask for the probability that the coordinate is exactly equal to one particular real number. A single point has measure zero, and the probability of one exact value is therefore zero.

What does make sense is the probability that the particle lies in an interval. Susskind temporarily





continues to use the symbol p for probability density, but to keep it separate from the later use of p for momentum, we will write the density as $P(x)$. The probability that the particle is found between x and $x + \Delta x$ is

$$P(x \in [x, x + \Delta x]) = \int_x^{x+\Delta x} P(x')\, dx'. \tag{4.4}$$

Accordingly, the normalization rule becomes

$$\int P(x)\, dx = 1. \tag{4.5}$$

This is the first replacement. In the discrete case the probabilities themselves add to 1; in the continuous case the area under the density curve adds to 1.

The surviving frame is strong evidence for the geometry of the argument. The lecture is not yet doing something sophisticated here. It is making a simple point carefully: for a continuous variable, the physically meaningful probability is associated with a small interval and therefore with a small area under the graph.

At this stage one replacement has been made: discrete probabilities have become probability densities. But a second replacement is now forced on us. If the basis is labeled by a continuous

variable, what replaces $\langle m|n\rangle = \delta_{nm}$? Before returning to probabilities, the lecture deliberately steps aside to introduce the Dirac delta.

4.3 The Dirac Delta and the Particle on a Line

The Dirac delta is introduced as the continuous analogue of the Kronecker delta. Susskind presents it first in the intuitive way and only then in the operational one. Intuitively, $\delta(x-y)$ is zero whenever $x \neq y$, but at $x = y$ it is concentrated into an infinitely narrow, infinitely tall spike of unit area:

$$\delta(x-y) = 0 \quad \text{for } x \neq y, \qquad \int \delta(x-y)\,dx = 1. \tag{4.6}$$

As he says, no honest ordinary function behaves this way. But the spike picture is a good guide to how the object works.

The more careful statement is that the delta is defined by what it does under integration:

$$\int \delta(x-y) f(x)\,dx = f(y). \tag{4.7}$$

Only the value of the function at $x = y$ survives. The heuristic picture and the operational definition are not rivals here. In the lecture they play different roles. The spike picture carries the intuition; the integral rule carries the mathematics.

With that in hand, we can now write the continuum analogue of orthogonality:

$$\langle x|y\rangle = \delta(x-y). \tag{4.8}$$

This is the natural replacement of $\langle m|n\rangle = \delta_{nm}$ when the basis is labeled by position. If $x \neq y$, the two position states are orthogonal; if $x = y$, the overlap is concentrated at one point, exactly as the delta function prescribes.

4.3.1 Question & Answer

Question. How can $\langle x|y\rangle$ be a delta function, and in what sense can a position eigenvector have infinite norm?

Answer. The lecture answers this by not pretending the continuum is innocent. Instead, it backs up to a dense discrete approximation. Imagine replacing the line by a set of lattice sites separated by a tiny spacing a, and let $|n\rangle$ be the corresponding orthonormal basis:

$$\langle n|m\rangle = \delta_{nm}.$$

Now define the continuum-normalized position state by stretching the discrete state:

$$|x\rangle = \frac{1}{\sqrt{a}}\,|n\rangle.$$

If x corresponds to site n and y to site m, then

$$\langle x|y\rangle = \frac{1}{a}\,\delta_{nm}. \tag{4.9}$$

This object is zero unless the two sites coincide. When they do coincide, its height is $1/a$. As a becomes very small, we get exactly the picture of a spike of width a and height $1/a$, that is, the Dirac delta.

So yes, the continuum-normalized position states have divergent norm. But that is not a defect; it is the normalization convention adapted to continuum integration. One almost never needs $\langle x|x\rangle$ by itself. What one actually uses is the delta function inside integrals, where it behaves perfectly well. The lecture's point is that the Dirac delta is not mysterious once one sees it as the rescaled limit of the Kronecker delta.

4.4 Born's Rule in the Continuum

Having inserted the delta-function detour, the lecture now returns, very explicitly, to the rules of probability. This return matters. Susskind is not content merely to quote $P(y) = |\psi(y)|^2$. He wants to derive it from the same postulates used in the discrete case.

The states are now wavefunctions, that is, complex functions of a real variable x, and the inner product on this vector space is

$$\langle\phi|\psi\rangle = \int \phi^*(x)\psi(x)\,dx. \tag{4.10}$$

A particle known to be localized at the point y is represented in the x-representation by the wavefunction $\delta(x-y)$. Therefore the amplitude to detect the particle at y in the state $|\psi\rangle$ is

$$\langle y|\psi\rangle = \int \delta(x-y)\psi(x)\,dx \tag{4.11}$$

$$= \psi(y). \tag{4.12}$$

This is the formal version of a statement that is often introduced heuristically: the wavefunction evaluated at y is the amplitude to find the particle at y.

Now we simply apply the probability postulate in its continuum form:

$$P(y) = |\langle y|\psi\rangle|^2. \tag{4.13}$$

Using the previous result,

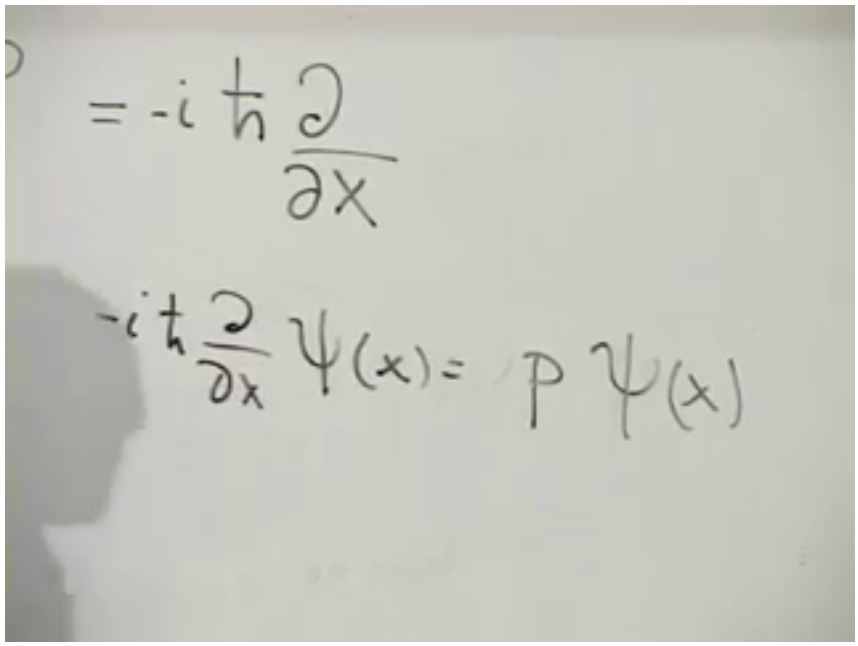
$$P(y) = |\psi(y)|^2 = \psi^*(y)\psi(y). \tag{4.14}$$

Susskind remarks that one could have begun with this formula directly. But the point of the lecture is different. The point is to show that nothing new has been added. We have simply taken the discrete postulates, replaced sums by integrals and Kronecker deltas by Dirac deltas, and then carried the same formal logic through.

The wavefunction of a particle on a line is therefore a probability amplitude in precisely the same sense that $\langle n|\psi\rangle$ is a probability amplitude in the discrete case. The difference is only that the label on the basis is now continuous.

4.5 Particle on a Circle: Periodicity, Momentum Eigenfunctions, and Quantization

The next transition in the lecture is from one system to another, not from one formalism to another. We now study a particle moving on a circle of radius R. To connect it with the line notation, the



circle is cut and laid out as the interval

$$0 \leq x \leq 2\pi R.$$

Everything looks the same as for a particle on a line until one remembers that this interval really came from a circle. That memory is encoded in one extra condition:

$$\psi(x + 2\pi R) = \psi(x). \tag{4.15}$$

The allowed state space is therefore the vector space of periodic complex functions on the interval. The lecture stops to note that this is indeed a vector space: multiplying a periodic function by any complex number keeps it periodic, and the sum of two periodic functions is periodic as well.

Now Susskind returns to momentum. In the x-representation, the momentum operator acts as

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}. \tag{4.16}$$

The frame below is the strongest board witness for the eigenvalue equation itself; the operator line above is only partly visible in the frame, but the transcript settles its standard form.

The momentum eigenvalue equation is

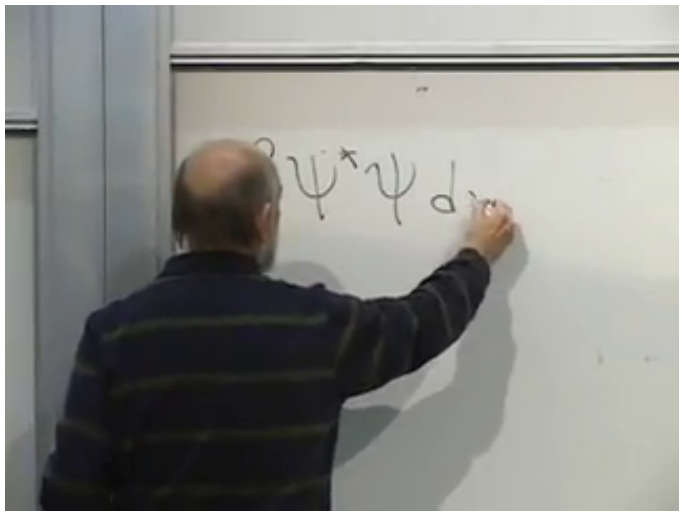
$$-i\hbar \frac{\partial}{\partial x}\psi(x) = p\,\psi(x). \tag{4.17}$$

Solving it is immediate:

$$\frac{d\psi}{dx} = \frac{ip}{\hbar}\,\psi, \tag{4.18}$$

$$\psi_p(x) = C\,e^{ipx/\hbar}. \tag{4.19}$$

So the momentum eigenfunctions are plane waves. This is one of the first places where the wave language of quantum mechanics becomes literal.



The lecture then normalizes the eigenfunctions on the circle.

The board shows the integrand $\psi^*\psi$ and the measure dx; the equality to 1 comes from the transcript rather than the frame. The normalization condition is

$$\int_0^{2\pi R} \psi_p^*(x)\psi_p(x)\,dx = 1. \tag{4.20}$$

Since $\psi_p(x) = Ce^{ipx/\hbar}$, we have $\psi_p^*\psi_p = |C|^2$. Therefore

$$1 = |C|^2 \int_0^{2\pi R} dx = 2\pi R\,|C|^2. \tag{4.21}$$

Hence

$$C = \frac{1}{\sqrt{2\pi R}} \tag{4.22}$$

up to an irrelevant phase, and the normalized eigenfunction is

$$\psi_p(x) = \frac{1}{\sqrt{2\pi R}}\,e^{ipx/\hbar}. \tag{4.23}$$

For different eigenvalues $p \neq q$, the corresponding eigenfunctions are orthogonal. One can check this directly, but the lecture prefers to emphasize the general reason: eigenvectors of a Hermitian operator with different eigenvalues are orthogonal. Thus

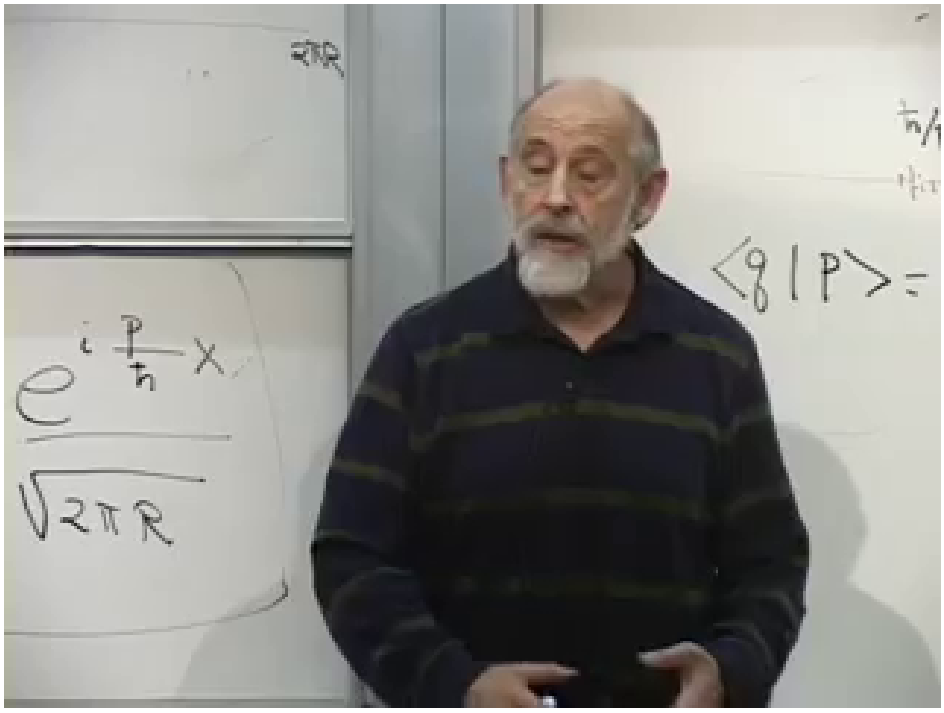
$$\int_0^{2\pi R} \psi_p^*(x)\psi_q(x)\,dx = 0 \qquad (p \neq q). \tag{4.24}$$

The next frame is less legible mathematically, but it is valuable because it preserves the board layout: the explicit plane-wave form on one side and the bra-ket representation on the other.

The exact bra-ket label in the screenshot is partly obscured, so the following is a cautious transcript-based reconstruction:

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi R}}\,e^{ipx/\hbar}. \tag{4.25}$$

That is, a momentum eigenvector, when written in the position basis, is a normalized plane wave.



Now comes the crucial new step. Not every plane wave is allowed, because the state must be periodic. Imposing periodicity gives

$$e^{ip(x+2\pi R)/\hbar} = e^{ipx/\hbar}. \tag{4.26}$$

Cancelling the common factor $e^{ipx/\hbar}$, we obtain

$$e^{i2\pi Rp/\hbar} = 1. \tag{4.27}$$

This happens precisely when the exponent is an integer multiple of $2\pi i$:

$$\frac{2\pi Rp}{\hbar} = 2\pi n, \qquad n \in \mathbb{Z}. \tag{4.28}$$

Therefore the allowed momenta are

$$p_n = \frac{n\hbar}{R}. \tag{4.29}$$

The position on the circle is continuous, but the momentum is discrete. The lecture insists on the origin of this discreteness: it comes from periodicity of the wavefunction.

Now let L be the angular momentum. Classically,

$$L = Rp. \tag{4.30}$$

Substituting the allowed momenta, we find

$$L_n = Rp_n = n\hbar. \tag{4.31}$$

The factor of R disappears. Angular momentum is quantized in units of $\hbar$, independent of the size of the circle.

4.5.1 Question & Answer

Question. Does the large-R limit give us classical mechanics, or only a dense momentum spectrum?

Answer. Only a dense spectrum. As $R \to \infty$, the spacing

$$\Delta p = \frac{\hbar}{R}$$

between neighboring momentum values goes to zero. The discrete spectrum becomes dense, and the lecture says that it is then convenient to pass from Kronecker-delta normalization to Dirac-delta normalization in momentum space. But nothing essentially classical has happened yet.

Classical mechanics would require that a particle could have a sharply defined position and a sharply defined momentum simultaneously. The large-R limit does not provide that. It merely makes the momentum eigenvalues closely packed. The real obstruction lies in the incompatibility of the position and momentum operators, which is the next topic.

4.6 Compatibility, Common Eigenvectors, and the Commutator $[x, p]$

The lecture now answers the classical-limit question by shifting from eigenvalues to states. A dense spectrum is not enough. The issue is whether the system can be in a state that is simultaneously an eigenstate of position and momentum.

Suppose two observables are represented by operators A and B. They are compatible if there exists a complete basis of vectors that are eigenvectors of both at once. Write

$$A|n\rangle = \alpha_n|n\rangle, \qquad B|n\rangle = \beta_n|n\rangle. \tag{4.32}$$

Then

$$AB|n\rangle = \alpha_n\beta_n|n\rangle, \qquad BA|n\rangle = \beta_n\alpha_n|n\rangle. \tag{4.33}$$

Since ordinary numbers commute, these two results agree. This leads to the criterion

$$[A, B] \equiv AB - BA = 0. \tag{4.34}$$

The lecture states, without proving it, that this is both necessary and sufficient: two observables are compatible precisely when their commutator vanishes.

Position and momentum do not satisfy this condition. Position eigenstates are sharply localized; momentum eigenstates are plane waves. On the circle, each momentum eigenstate has constant magnitude around the entire circle; on the infinite line, it is delocalized over the whole axis. So the lecture's physical point is immediate even before the algebra: these are not the same kind of states, and there are no common eigenvectors.

The algebra confirms this. Let x act by multiplication and p act as $-i\hbar\, d/dx$. To determine the commutator, apply it to an arbitrary wavefunction $\psi(x)$:

$$(xp - px)\psi(x) = x\left(-i\hbar\frac{d\psi}{dx}\right) - \left(-i\hbar\frac{d}{dx}\right)(x\psi(x)) \tag{4.35}$$

$$= -i\hbar\, x\frac{d\psi}{dx} + i\hbar\,\frac{d}{dx}(x\psi(x)). \tag{4.36}$$

Now use the product rule,

$$\frac{d}{dx}(x\psi(x)) = \psi(x) + x\frac{d\psi}{dx}, \tag{4.37}$$

so that

$$(xp - px)\psi(x) = -i\hbar\, x\frac{d\psi}{dx} + i\hbar\left(\psi(x) + x\frac{d\psi}{dx}\right) \tag{4.38}$$

$$= i\hbar\,\psi(x). \tag{4.39}$$

Because this is true for every $\psi(x)$, we conclude that

$$[x, p] = xp - px = i\hbar. \tag{4.40}$$

This is the formal hinge of the lecture. As a statement about ordinary numbers it would be nonsense, because $xp = px$. But as a statement about operators it is exact. Susskind emphasizes that the right-hand side is just a number, and that is about as strong a failure of commutativity as one could have. There are no common eigenvectors at all.

The lecture also addresses a natural confusion here. Multiplication by x does not turn an arbitrary function into a constant multiple of itself. That is why generic functions are not eigenfunctions of the position operator. Only delta functions have the property that multiplication by x picks out one specific numerical value.

4.7 Two-Slit Interference as Superposition of States

The lecture now cashes out all of this formalism in the two-slit experiment. The central idea is not diffraction theory in detail. It is superposition. Quantum states can be added, and the probability is computed only after the amplitudes have been added.

Start with one hole open. The wave emerging from that hole spreads out and reaches the screen with a varying phase. Over a small region of the screen one may approximate the vertical dependence by something like e^{ipy}, but Susskind is careful not to pretend that this is globally exact. A better heuristic description is

$$\psi_{\text{1-hole}}(y) \sim e^{i\varphi(y)}\rho(y), \tag{4.41}$$

where $\rho(y)$ is a slowly varying envelope and $\varphi(y)$ is a rapidly varying phase. The lecture also gives a more geometric outgoing-wave model:

$$\psi(r) \sim \frac{e^{ipr}}{r}, \qquad r = \sqrt{\ell^2 + y^2}. \tag{4.42}$$

The point of this formula is not to launch a full diffraction calculation. It is to explain why the local vertical frequency varies with y. The phase is pr, not simply py, so as one moves away from the center the apparent oscillation in the vertical direction speeds up. In physical language, the particle must have picked up more vertical momentum.

4.7.1 Question & Answer

Question. Why does one hole give a smooth blob, while two holes can create dark fringes where no particle ever lands?

Answer. For one hole, the phase may oscillate rapidly, but the probability is computed by multiplying the amplitude by its complex conjugate:

$$|\psi_{\text{1-hole}}(y)|^2 \sim \big(e^{i\varphi(y)}\rho(y)\big)\big(e^{-i\varphi(y)}\rho(y)\big) = \rho(y)^2. \tag{4.43}$$

The oscillating phase cancels. What remains is the smooth envelope. That is why a single hole can produce a broad blob even though the amplitude itself carries rapid phase variation.

With two holes open, however, the total state is the sum of two amplitudes:

$$\psi = \psi_1 + \psi_2.$$

When we square that sum, cross-terms survive. Those cross-terms contain the relative phase, and that relative phase produces the interference pattern. So one hole gives a smooth probability because its phase disappears in $|\psi|^2$; two holes give fringes because the relative phase does not disappear in $|\psi_1 + \psi_2|^2$.

To see the algebra in the form used in the lecture, let us work locally on the screen and choose units in which $\hbar = 1$. Approximate the two contributions by

$$\psi_1(y) \approx e^{ipy}, \qquad \psi_2(y) \approx e^{iqy}. \tag{4.44}$$

Each one separately has constant magnitude. Hence each one separately would give a uniform probability over that small interval. But with both holes open,

$$|\psi_1 + \psi_2|^2 = (e^{ipy} + e^{iqy})(e^{-ipy} + e^{-iqy}) \tag{4.45}$$

$$= 1 + 1 + e^{i(p-q)y} + e^{i(q-p)y} \tag{4.46}$$

$$= 2 + 2\cos\big((p-q)y\big). \tag{4.47}$$

This is not constant. It oscillates, and for those values of y where the cosine is -1, the probability is zero. These are the dark fringes.

That is the lecture's real payoff. Open one hole and a particle can arrive at a given point. Open the other hole and it can arrive there as well. Open both holes together and there are points that no particle reaches in the idealized limit. This is not because probabilities were added incorrectly; it is because amplitudes were added first, and only then squared.

The lecture repeatedly returns to the particle character of the phenomenon. One sends particles through one at a time. Each event is just a spot on the screen. But after many repetitions the spots build up the probability distribution. The distribution is always nonnegative, because it comes from an amplitude times its complex conjugate, but it need not be strictly positive. In the ideal limit it can vanish.

The dependence on slit separation now reads like ordinary beats. If the two holes are very close together, the local vertical frequencies p and q are almost the same, so $p - q$ is small and the cosine varies slowly. The fringe spacing is therefore large. If the holes are farther apart, $p - q$ is larger and the fringes become more closely spaced.

The lecture also adds a useful caution. A single aperture with sharp edges can itself produce diffraction structure because each edge behaves as a secondary source. That is a different effect. The point here is the distinct and stronger interference that arises from the superposition of the states associated with two separate holes.

Finally, the lecture stops at exactly the right place. It refuses, for the time being, to answer the which-path detector question. If a detector tells us which slit the particle went through, the interference pattern is destroyed, but to explain that properly one needs composite systems and entanglement. The same is true of the reversibility question that comes up at the end. Those topics are deferred until Hamiltonians, evolution, and tensor-product structure are in place.

4.8 Summary

The lecture unfolds as one deliberate chain of replacements and payoffs. We begin with the discrete case: probabilities $P(n)$, orthonormal basis states, amplitudes, and the Kronecker delta. We then pass to continuous variables: exact point-probabilities give way to interval probabilities, sums become integrals, and the Kronecker delta becomes the Dirac delta. With those replacements in hand, the continuum Born rule follows from the same abstract postulates:

$$P(y) = \psi^*(y)\psi(y).$$

The move from the line to the circle changes not the formalism but the allowed state space, and periodicity then forces momentum and angular momentum to be quantized. The student's natural question about the large-R limit pushes the lecture to the deeper issue of compatibility, culminating in the operator identity

$$[x, p] = i\hbar.$$

That identity explains why position and momentum are not simultaneously sharp. The two-slit experiment then gives the physical payoff: amplitudes add, probabilities do not, and the cross-term in

$$|\psi_1 + \psi_2|^2$$

creates an interference pattern with genuine dark fringes. The lecture ends by leaving the detector question open, not because it is unimportant, but because it belongs to the next layer of the subject.